CHAPTER

22

Cerebellum

The cerebellum occupies the posterior cranial fossa, separated from the occipital lobes of the cerebral hemispheres by the tentorium cerebelli. It is the largest part of the hindbrain; in adults, the weight ratio of cerebellum to cerebrum is approximately 1 : 10, and in infants 1 : 20. The cerebellum lies dorsal to the pons and medulla, from which it is separated by the fourth ventricle. It is joined to the brainstem by three bilaterally paired cerebellar peduncles.

The basic internal organization of the cerebellum is of a superficial cortex overlying a core of white matter. The cortex is highly convoluted, forming lobes and lobules that are further subdivided into folia (leaflets), separated by intervening transverse fissures. Aggregations of neuronal cell bodies embedded within the white matter form the fastigial (medial), globose (posterior interposed), emboliform (anterior interposed) and dentate (lateral) nuclei, which are collectively known as the (deep) cerebellar nuclei.

The cerebellum may be subdivided into a number of modules, each consisting of a longitudinal cortical zone, a cerebellar or vestibular target nucleus, and a supporting olivocerebellar climbing fibre system. Apart from their connections, the longitudinal cortical zones are characterized by their immunohistochemical properties. The cerebellum receives input from peripheral receptors and from motor centres in the spinal cord and brainstem and from large parts of the cerebral cortex through two different afferent systems: mossy and climbing fibres. It is located as a side path to the main ascending sensory and descending motor systems, and it functions to coordinate movement. During movement, the cerebellum provides corrections that are the basis for precision and accuracy, and it is critically involved in motor learning and reflex modification. Cerebellar output is directed to the thalamus and from there to the cerebral cortex, and also to brainstem centres such as the red nucleus, vestibular nuclei and reticular nuclei, which themselves give rise to descending spinal pathways.

Ideas on the involvement of the cerebellum in motor functions were derived mainly from movement disorders seen in experimental studies, summarized by Luciani (1891) in his triad of atonia, astasia and asthenia, and in human patients with cerebellar lesions who displayed the well-known symptoms of gait disturbances, limb ataxia, dysmetria, atonia and eye movement disorders (Glickstein et al 2009). Latterly, the observation that lesions of the cerebellar hemisphere not only resulted in minor and transient motor symptoms but also induced a cerebellar cognitive/affective syndrome (Schmahmann 2004) prompted the suggestion that the human cerebellum is also concerned with non-motor functions. These conceptual developments went hand in hand with the use of more sensitive experimental methods to trace cerebellar connectivity, mainly in subhuman primates, and the application of modern imaging techniques to the human brain. Although it is now recognized that the cortex is more heterogeneous than previously supposed, and despite our extensive knowledge of the sphere of influence of the cerebellum and the microcircuitry of its cortex and nuclei, we still do not fully understand how it contributes to motor and non-motor systems. The observation by Thomas Willis (1681) that 'the Spirits inhabiting the Cerebel perform unperceivedly and silently their Work of Nature without our Knowledge or Care' remains true today.

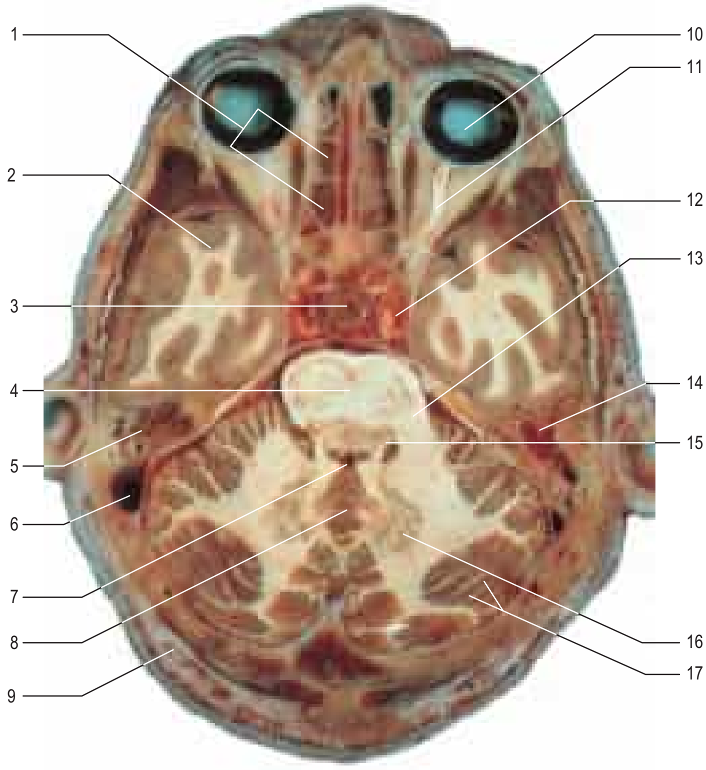

Fig. 22.1 A horizontal section through the cerebellum and brainstem. 1. Ethmoidal air cells. 2. Temporal lobe of brain. 3. Hypophysis. 4. Pons. 5. Cochlea. 6. Sigmoid sinus. 7. Fourth ventricle. 8. Vermis. 9. Diploë of occipital bone. 10. Eyeball. 11. Optic nerve. 12. Internal carotid artery. 13. Middle cerebellar peduncle. 14. Petrous temporal bone. 15. Superior cerebellar peduncle. 16. Dentate nucleus. 17. Folia of cerebellar cortex. (Courtesy of Dr GJA Maart.)

EXTERNAL FEATURES AND RELATIONS

The cerebellum consists of two large, laterally located hemispheres that are united by a midline vermis (**Figs 22.1–22.3**). Numerous sulci and fissures of varying depth subdivide it into lobes, lobules and folia (leaflets) (**Figs 22.4–22.5**). The primary fissure, the deepest fissure on a sagittal section, divides it into anterior and posterior lobes. Paramedian fissures, shallow in the anterior cerebellum but prominent more posteriorly, separate the vermis from the cerebellar hemispheres. Both the anterior and posterior vermis and hemisphere are subdivided into lobules that received their names in the eighteenth and nineteenth centuries from their shape, their position or their likeness to anatomical structures in other body parts (Glickstein et al 2009). This classical nomenclature (see Fig. 22.4, right panel) was largely replaced in the early twentieth century by a nomenclature based on Bolk's (1906) comparative anatomical investigations (see Fig. 22.4, left panel).

Bolk distinguished the relatively independent 'folial chains' of the vermis and the hemispheres. In later studies, this relative independence was found to reflect the continuity or discontinuity of the cortex between the lobules within a folial chain, or between the folial chains of the vermis and the hemispheres. Bolk used the cerebellum of a small lemur for his initial description (see Fig. 22.5F–G) and summarized the configuration of the folial chains in a stick diagram (see Fig. 22.5H). His description proved to be applicable to the cerebella of all the mammals he studied, including the human cerebellum.

Larsell (Larsell and Jansen 1972) based his subdivision of the cerebellum on embryological studies of the emergence of the transverse fissures with time. Contrary to Bolk, who emphasized the continuity within the folial chains, Larsell attached great importance to the mediolateral continuity of the lobules of the vermis and the hemispheres, and distinguished 10 lobules in the cerebellum, indicated using Roman numerals I–X for the vermis and the prefix H for the hemisphere. The correspondence of Larsell's lobules with the classical nomenclature is shown in Figure 22.4.

Lobules (H)I–V constitute the anterior lobe. Lobule I, the lingula, is conjoined with the superior medullary velum. Lobules VI (declive) and

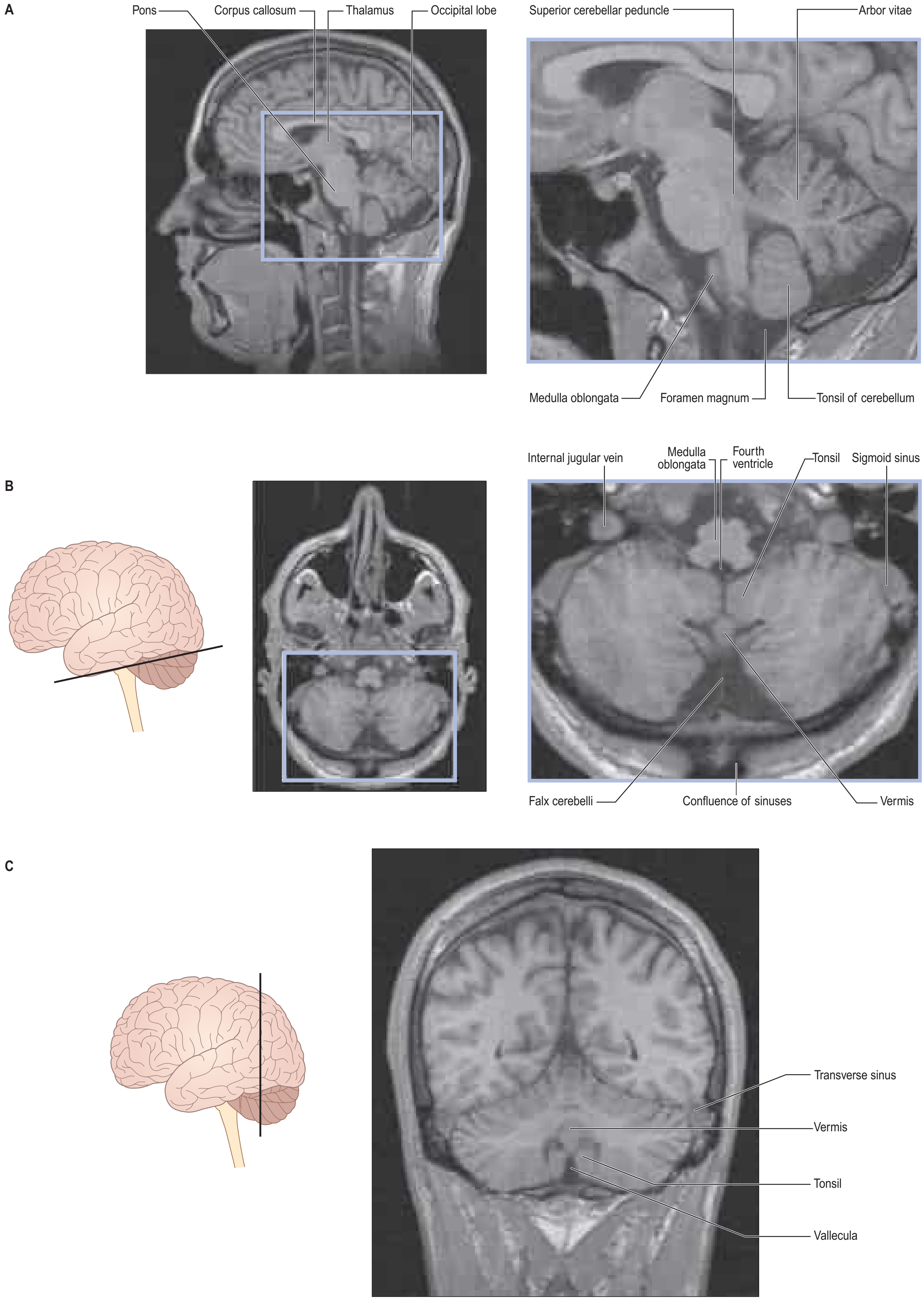

Fig. 22.2 Magnetic resonance images of the cerebellum of a 16-year-old female. **A**, Sagittal view. **B**, Axial view. **C**, Coronal view. (Courtesy of Drs JP Finn and T Parrish, Northwestern University School of Medicine, Chicago.)

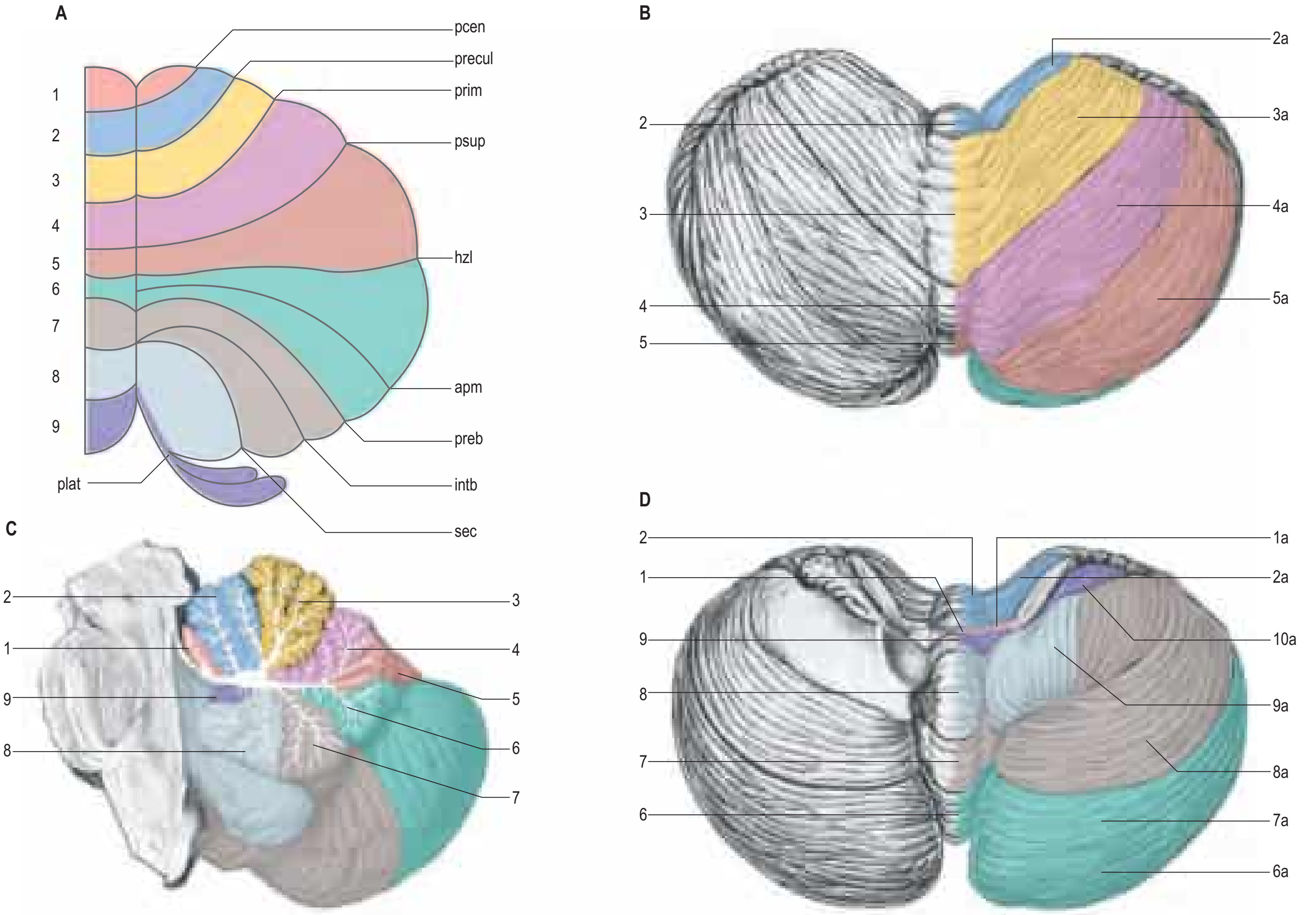

Fig. 22.3 The terminology of the cerebellar lobes and fissures, using a schematic 'unrolled' diagram as a frame of reference. **A**, Unrolled cerebellar cortex. The lobules are labelled by numbers and the fissures between the lobules are listed. **B**, The cerebellum viewed from above. **C**, A median sagittal section of cerebellum. The lobules are numbered and listed. **D**, The cerebellum viewed from below. Key and abbreviations: *Anterior lobe*: 1, lingula; 2, central; 3, culmen, vermis. *Posterior lobe*: 4, declive; 5, folium; 6, tuber; 7, pyramis; 8, uvula; 9, nodule. *Fissures*: apm, ansoparamedian; hzl, horizontal; intb, intrabiventral; pcen, precentral; plat, posterolateral; preb, prebiventral; precul, preculminate; prim, primary; psup, posterior superior; sec, secondary. *Hemisphere*: 1a, wing of lingula; 2a, wing of central lobule; 3a, anterior quadrangular lobule; 4a, posterior quadrangular lobule; 5a, superior semilunar lobule; 6a, inferior semilunar lobule; 7a, gracile lobule; 8a, biventral lobule; 9a, tonsil of cerebellum, 10a, flocculus.

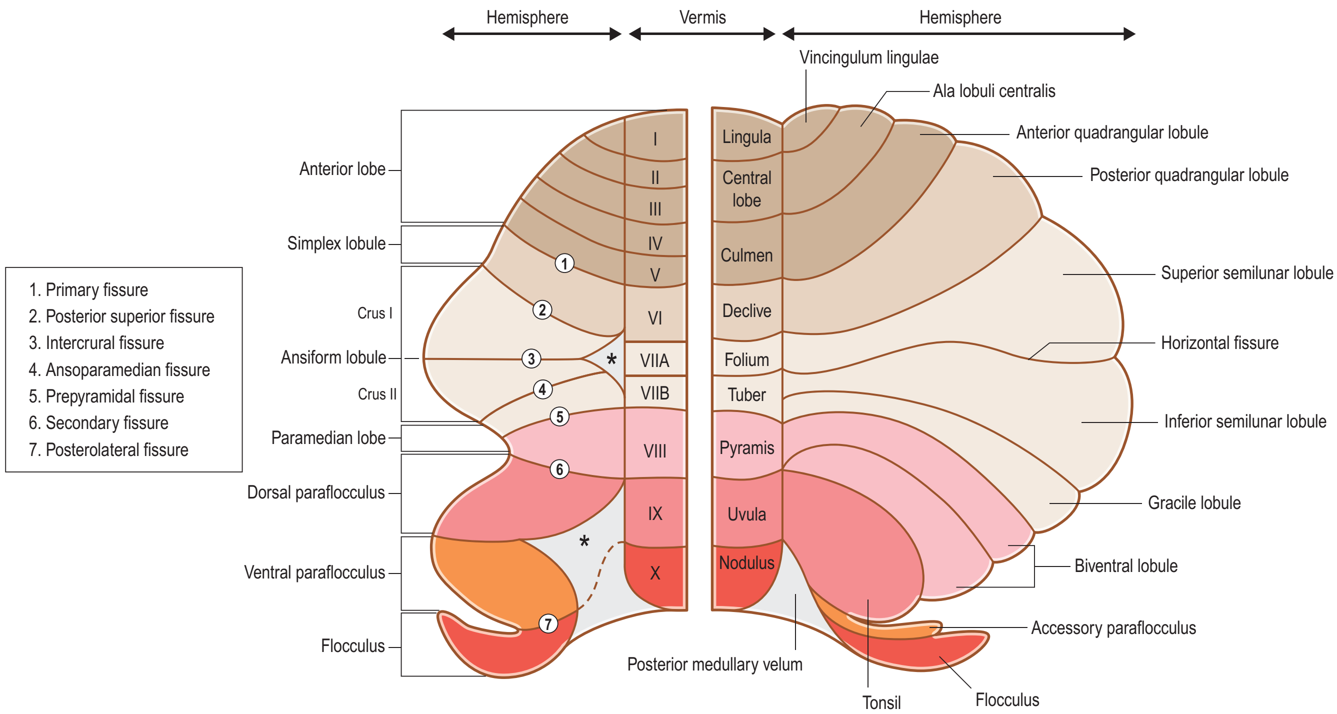

Fig. 22.4 Cerebellar nomenclature. The left-hand panel illustrates the comparative anatomical nomenclature for the hemisphere and Larsell's numbering system for the lobules of the vermis (Larsell and Jansen 1972). The right-hand panel shows the classical nomenclature. The homology of these lobules is indicated using the same colour. Asterisks denote areas devoid of cortex in the centre of the folial rosettes of the ansiform lobule and the paraflocculus.

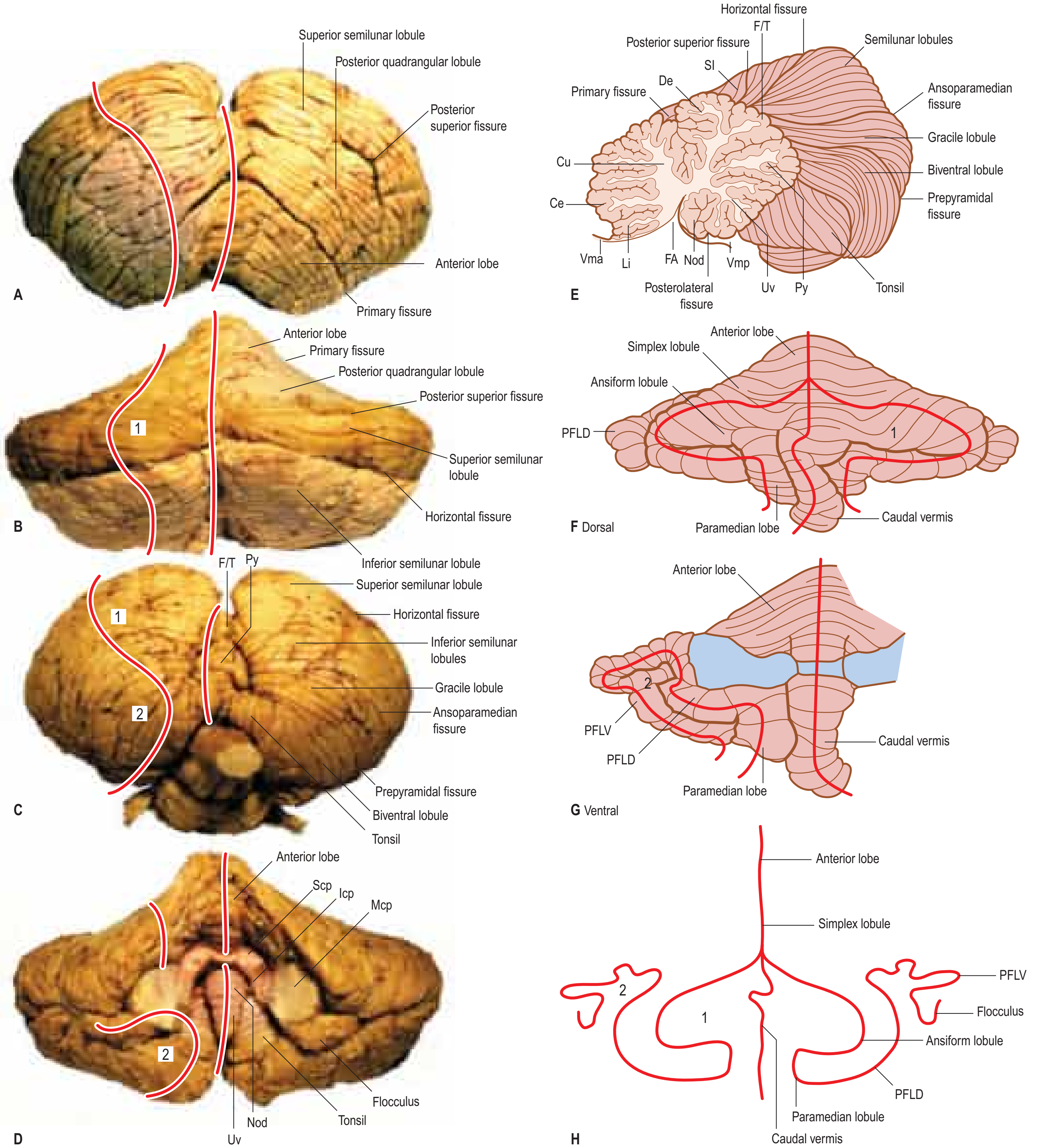

Fig. 22.5 A–D, Anterior, dorsal, posterior and ventral views of the human cerebellum. **E**, A sagittal section of the human cerebellum. **F–G**, Dorsal and ventral views of the cerebellum of *Lemur albifrons*, Bolk's (1906) prototype for his ground plan of the mammalian cerebellum. Two loops are present in the folial chain of the hemisphere: (1) as the ansiform lobule, (2) as the paraflocculus. The course of the folial chains of the vermis and hemisphere in A–D and F–G is indicated with red lines. **H**, Bolk's stick diagram of the folial chains of the vermis and hemisphere. Key and abbreviations: 1, 2, Ansiform and parafloccular loops of the folial chain of the hemisphere; Ce, central lobule; Cu, culmen; De, declive; FA, fastigium; F/T, folium and tuber; Icp, inferior cerebellar peduncle; Li, lingula; Mcp, middle cerebellar peduncle; Nod, nodulus; PFLD, dorsal paraflocculus; PFLV, ventral paraflocculus; Py, pyramis; Scp, superior cerebellar peduncle; SI, simplex (posterior quadrangular) lobule; Uv, uvula; Vma, anterior (superior) medullary velum; Vmp, posterior medullary velum.

HVI (posterior quadrangular lobule) are also known as Bolk's simplex lobule. Behind the primary fissure, the folium (lobule VIIA) and tuber vermis (VIIB) are separated by the deep paramedian fissure from the superior semilunar lobule (HVIIA), the inferior semilunar lobule and the gracile lobule (together corresponding to HVIIA). Superior and inferior semilunar lobules correspond to the crus I and II of Bolk's ansiform lobule. Their folia fan out from the deep horizontal fissure that represents the intercrural fissure. The gracile lobule corresponds to the rostral part of Bolk's paramedian lobule. Its caudal portion is formed by the biventral lobule (HVIII), the hemisphere from the pyramis (VIII). Lobule VIII (the pyramis) is continuous with the biventral lobule (HVIII) laterally. The gracile lobule corresponds to the rostral part of Bolk's paramedian lobule; the biventral lobule corresponds to its caudal portion. The tonsil (HIX) corresponds to the dorsal paraflocculus in the monkey. In the human, the folial loop of the tonsil is directed medially, contrary to the situation in most mammals, where the paraflocculus arches laterally. The flocculus appears as a double folial rosette; its dorsal leaf is known as the accessory paraflocculus of

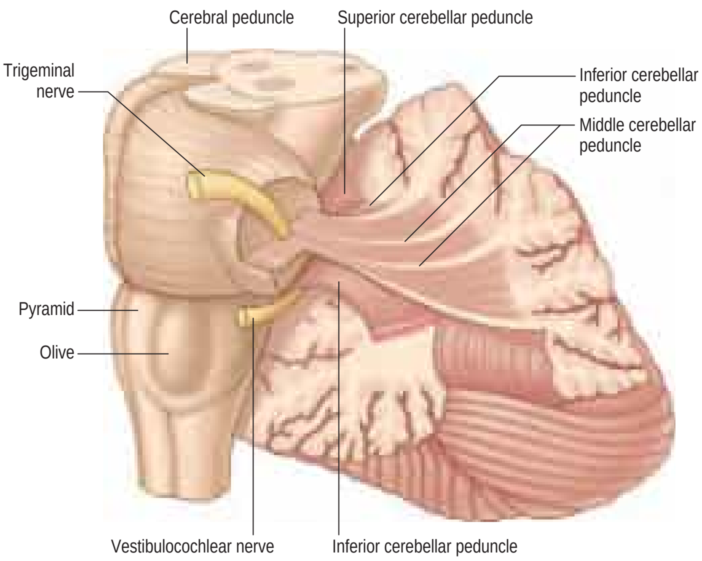

Fig. 22.6 Dissection of the left cerebellar hemisphere and its peduncles.

Henle, while its ventral leaf represents the true flocculus. The accessory paraflocculus corresponds to the ventral paraflocculus in the monkey. Both these lobules belong to the vestibulocerebellum. The cortex between the tonsil and the accessory paraflocculus is interrupted. Between lobule X (the nodulus) and the flocculus (HX) with the accessory paraflocculus, the cortex is absent and the tissue is stretched out as the inferior medullary velum.

Two magnetic resonance imaging (MRI) atlases of the cerebellum have been published to aid localization in functional MRI (fMRI) (Schmahmann et al 1999, Diedrichsen 2006). The authors use Larsell's numerals and retain Bolk's terms crus I and II, but discard Larsell's use of the prefix H for the lobules of the hemisphere. As a consequence, it is difficult to determine whether descriptions of lobules using these criteria refer to the vermis or to the hemisphere.

CEREBELLAR PEDUNCLES

Three pairs of peduncles connect the cerebellum with the brainstem (**Fig. 22.6**; see also Fig. 21.19).

The middle cerebellar peduncle is the most lateral and by far the largest of the three. It passes obliquely from the basal pons to the cerebellum and contains the massive pontocerebellar mossy fibre pathway, which is composed almost entirely of fibres that arise from the contralateral basal pontine nuclei, with a small addition from nuclei in the pontine tegmentum.

The inferior cerebellar peduncle is located medial to the middle peduncle. It consists of an outer, compact fibre tract – the restiform body – and a medial, juxtarestiform body. The restiform body is a purely afferent system; it receives spinocerebellar fibres and the trigeminocerebellar, cuneocerebellar, reticulocerebellar and olivocerebellar tracts from the medulla oblongata (see Fig. 21.19). The juxtarestiform body is mainly an efferent system, made up almost entirely of efferent Purkinje cell axons destined for the vestibular nuclei and uncrossed fibres from the fastigial nucleus. It also contains primary afferent mossy fibres from the vestibular nerve and secondary afferent fibres from the vestibular nuclei. The crossed fibres from the fastigial nucleus pass dorsal to the superior cerebellar peduncle as the uncinate tract, and enter the brainstem at the border of the juxtarestiform and restiform bodies.

The superior cerebellar peduncle contains all of the efferent fibres from the dentate, emboliform and globose nuclei, and a small fascicle from the fastigial nucleus. Its fibres decussate in the caudal mesencephalon, and are destined to synapse in the contralateral red nucleus and thalamus. The ventral spinocerebellar tract reaches the upper part of the pontine tegmentum, looping around the entrance of the trigeminal nerve to join this peduncle and unite with the spinocerebellar fibres entering through the restiform body.

INTERNAL ORGANIZATION

The vast majority of cerebellar neuronal cell bodies are located within the outer, highly convoluted cortical layer. Beneath the cortex, the cerebellar white matter forms an extensive central core, from which a characteristic branching pattern of nerve fibres (arbor vitae) extends towards the cortical surface (see Fig. 22.2). The white matter consists of afferent and efferent fibres travelling to and from the cerebellar cortex. Fibres cross the midline in the white core of the cerebellum and the superior medullary velum, effectively constituting a cerebellar 'commissure'.

CEREBELLAR CORTEX

Although the human cerebellum makes up approximately one-tenth of the entire brain by weight, the surface area of the cerebellar cortex, if unfolded, would be about half that of the cerebral cortex. The great majority of cerebellar neurones are small granule cells, so densely packed that the cerebellar cortex contains many more neurones than the cerebral cortex. Unlike the cerebral cortex, where a large number of diverse cell types are arranged differently in different regions, the cerebellar cortex contains a relatively small number of different cell types, which are interconnected in a highly stereotyped way.

The elements of the cerebellar cortex possess a precise geometric order, arrayed relative to the tangential, longitudinal and transverse planes in individual folia. Three layers are distinguished in the cerebellar cortex (**Figs 22.7**–**22.8**). A monolayer of large neurones with apical dendrites, first identified by Purkinje (Glickstein et al 2009), separates a layer of small granule cells from the superficial, cell-poor molecular layer. The Purkinje cell layer contains the large, pear-shaped somata of the Purkinje cells and the smaller somata of Bergmann glia. Clumps of granule cells and occasional Golgi cells penetrate between the Purkinje cell somata. The granular layer consists of the somata of granule cells and the initial segments of their axons; dendrites of granule cells; branching terminal axons of afferent mossy fibres; climbing fibres passing through the granular layer *en route* to the molecular layer; and the somata, basal dendrites and complex axonal ramifications of Golgi neurones.

The molecular layer contains a sparse population of neurones, dendritic arborizations, unmyelinated axons and radial fibres of neuroglial cells.

Purkinje cells

Purkinje cells are the only output neurones of the cortex. They are arranged in a single layer between the molecular and granular layers, and have a specific geometry that is conserved in all vertebrate classes (**Fig. 22.9**).

Their dendritic trees are flattened and orientated perpendicular to the parallel fibres in a plane transverse to the long axes of the folia (see Figs 22.7–22.8; see also Figs 3.3, 3.6). Large primary dendrites arise from the outer pole of a Purkinje cell. The proximal dendritic branches are smooth and are contacted by climbing fibres. The distal branches carry a dense array of dendritic spines (spiny branchlets) that receive synapses from the terminals of parallel fibres. Inhibitory synapses are also received from basket and stellate cells, and from the recurrent collaterals of Purkinje cell axons that contact the shafts of the proximal dendrites. The total number of dendritic spines per Purkinje neurone is in the order of 180,000. The axon of a Purkinje cell leaves the inner pole of the soma and crosses the granular layer to enter the subjacent white matter. The initial axon segment receives axo-axonal synaptic contacts from distal branches of basket cell axons. Beyond the initial segment, the axon enlarges, becomes myelinated and gives off collateral branches. The main axon ultimately terminates in one of the cerebellar or vestibular nuclei; recurrent axonal collaterals form a sagittally orientated plexus with terminations on neighbouring Purkinje cells and Golgi cells. Purkinje cells are inhibitory and use γ-aminobutyric acid (GABA) as their neurotransmitter.

Cortical interneurones

The cerebellar cortical interneurones were described by Ramón y Cajal (1906) (see Fig. 22.8). They can be divided into the interneurones of the molecular layer, the stellate and basket cells, and the Golgi cells of the granular layer. All interneurones are inhibitory. Those of the molecular layer use GABA as their neurotransmitter. Most Golgi cells are glycinergic. Stellate cells are located in the upper molecular layer, their axons terminating on Purkinje cell dendrites. Basket cells occupy the deep molecular layer, their axons terminating on a series of Purkinje cells with baskets surrounding their somata, ending in a plume around their initial axon. The dendrites of these interneurones and their axons are oriented in the sagittal plane. Golgi cell dendrites are located in the

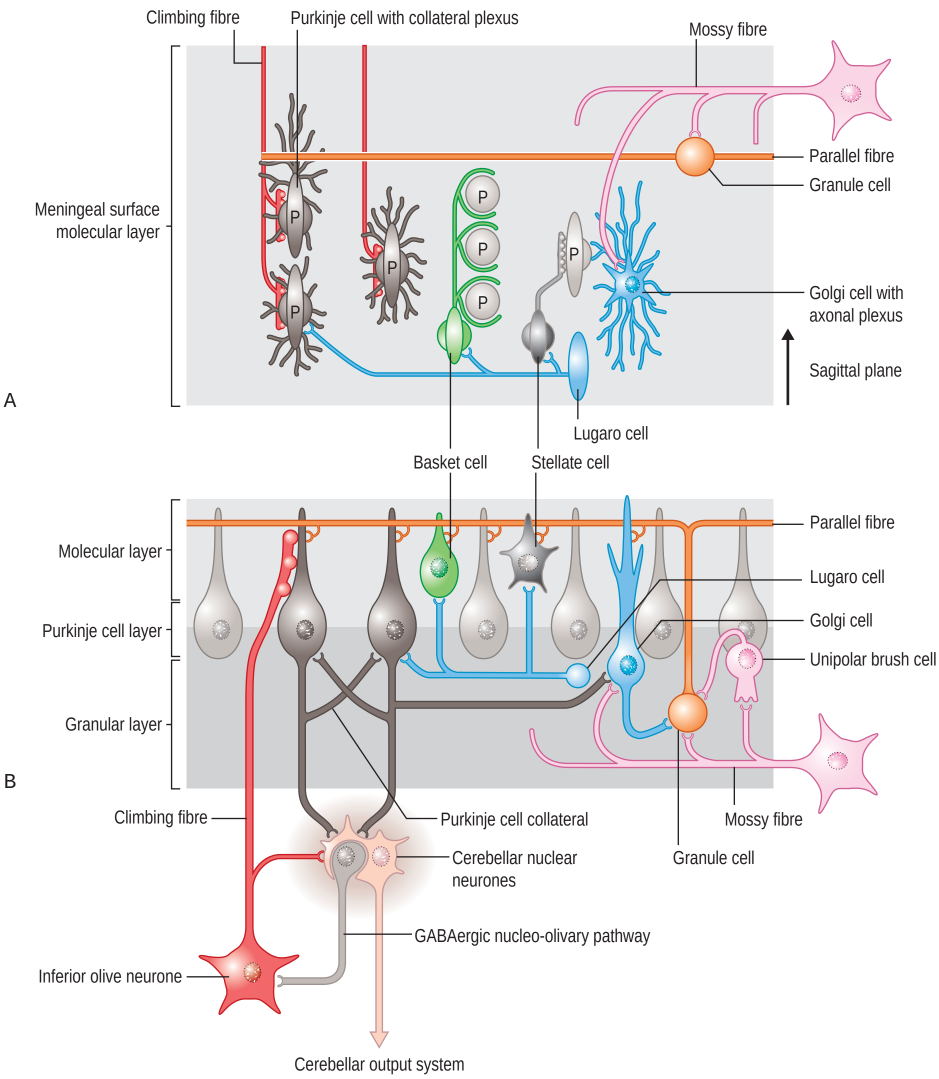

Fig. 22.7 The circuitry of the cerebellar cortex. Glutamatergic neurones are shown in dark grey, GABAergic neurones in red and glycinergic elements in blue. **A**, A transparent dorsal view of the cortex showing the orientation of its major elements. Dendrites of Purkinje, stellate and basket cells, the collateral plexus of Purkinje cells, the cell bodies of the Lugaro cells, and the climbing fibres and the axonal plexus of the Golgi cells are orientated in the sagittal plane. **B**, A transverse section. Parallel fibres and the axons of the Lugaro cells are the only elements with a transverse orientation.

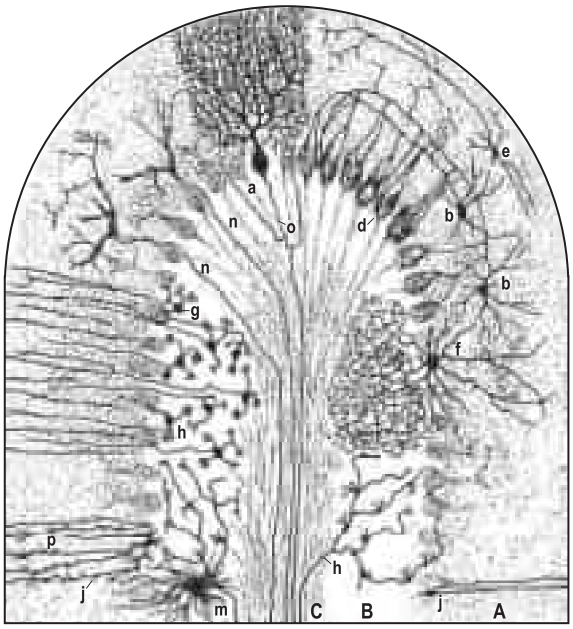

Fig. 22.8 A sagittal section through a cerebellar folium showing the different cell types of the cerebellar cortex. Abbreviations: A, molecular layer; a, Purkinje cell; B, granular layer; b, basket cells; C, white matter; d, baskets of basket cell; e, stellate cell; f, Golgi cell; g, granule cell with ascending axon; h, mossy fibre; m, astrocyte; n, climbing fibre; o, Purkinje cell axon with collaterals; j, p, Bergmann glial cells. (Redrawn from Cajal SR y. 1906 Histologie du système nerveux de l'homme et des vertebrés. Maloine, Paris.)

granular and molecular layers. Their axonal plexus occupies the granular layer, where it terminates on the granule cell dendrites, and also has its greatest dimension in the sagittal plane (see Fig. 22.7). Golgi cells are innervated by collaterals of mossy fibres and Purkinje cell axons. The dendrites of interneurones in the molecular layer are contacted by the parallel fibres. Synaptic contacts between climbing fibres and the dendrites or cell bodies of cerebellar interneurones in the molecular or granular layers have not been observed. However, interneurones of the molecular layer can be activated by 'spillover' of glutamate from the climbing fibres (Galliano et al 2013, Szapiro and Barbour 2007). Golgi cells, therefore, provide feed-back inhibition to the granule cells. Interneurones of the molecular layer provide feed-forward inhibition to the Purkinje cells. Stellate cells (Mann-Metzer and Yarom 2000) and Golgi cells (Dugué et al 2009) are electrotonically coupled. The extent of this coupling is not known; it may be restricted to the sagittal compartments that are one of the main features of the connectivity of the cerebellum, discussed below.

Two other types of interneurone exist. Lugaro cells are cigar-shaped neurones located at the level of the Purkinje cells (Lainé and Axelrad 1996) (see Fig. 22.7). These glycinergic neurones innervate the stellate and basket cells, and provide a long, transversely orientated axon that terminates on Purkinje cells. They receive a strong input from an extra-cerebellar serotoninergic system. Monopolar brush cells are excitatory neurones, mainly found in vestibular-dominated regions of the cerebellum (Mugnaini et al 1997), where they are considered to be a 'booster' system for vestibular mossy fibre input. Mossy fibres terminate with extremely large synapses on the base (the 'brush') of these cells. Their axons terminate as mossy fibres on the granule cells.

CEREBELLAR NUCLEI

The four cerebellar nuclei were first described by Stilling (1864) as comprising (from medial to lateral) the fastigial nucleus, the emboliform and globose nuclei, and the dentate nucleus (Fig. 22.10). The

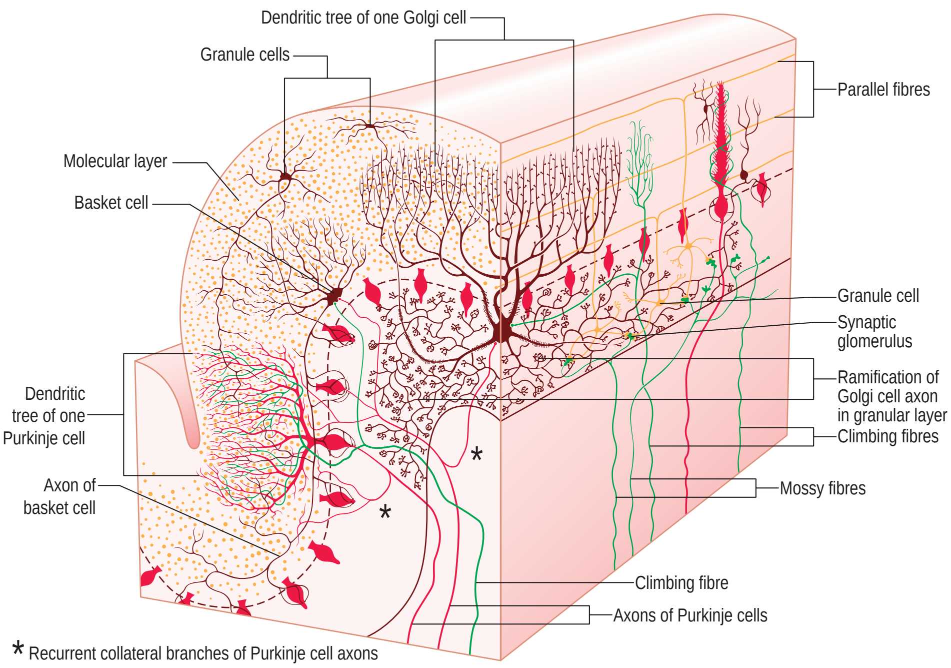

Fig. 22.9 The general organization of the cerebellar cortex. A single folium has been sectioned vertically, both in its longitudinal axis (right side of diagram) and transversely.

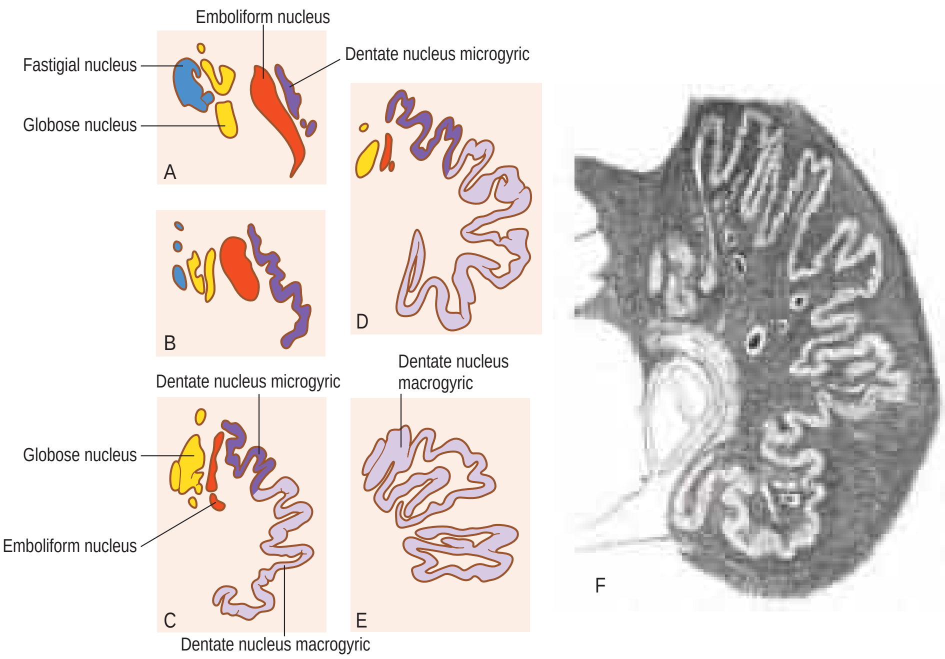

Fig. 22.10 The human cerebellar nuclei. **A–E**, Transverse sections through the cerebellar nuclei, A being the most rostral level. The dentate nucleus can be subdivided into dorsomedial microgyric and ventrocaudal macrogyric parts. **F**, A Weigert-stained section through the dentate nucleus, showing its subdivision into micro- and macrogyric parts. (A–E Redrawn from Larsell O, Jansen J 1972 The comparative anatomy and histology of the cerebellum. III. The human cerebellum, cerebellar connections, and cerebellar cortex. Minneapolis, University of Minnesota Press. F, Reproduced with permission from Winkler C 1926 De bouw van het zenuwstelsel. Haarlem, de erven Bohn.)

emboliform and globose nuclei are also known as the anterior and posterior interposed nuclei. The nuclei form two interconnected groups: a rostrolateral group consisting of the emboliform and dentate nuclei, and a caudomedial group including the fastigial and globose nuclei.

A collection of small, cholinergic neurones extends from the flocculus to the nodulus in the roof of the fourth ventricle, invading the spaces between the nuclei. These cells are known as the basal interstitial nucleus (Langer 1985); their connections are not known.

The dentate nucleus is located most laterally and is by far the largest of the group. It has the shape of a crumpled purse; the main efferent pathway of the cerebellum, the brachium conjunctivum, emerges from its hilus. The convolutions of the dentate nucleus are narrow rostromedially and much wider ventrocaudally. Interestingly, these micro- and macrogyral characteristics of the human dentate were observed by Vicq-d'Azir, who coined its name in the eighteenth century (Glickstein et al 2009). Recently, rostromedial motor and ventrocaudal non-motor divisions have been distinguished in the human dentate nucleus using fMRI (Küper et al 2012). Their significance and the possible correspondence with the anatomical subdivisions of the dentate are considered below.

The cerebellar nuclei contain cells of all sizes. Glutamatergic relay neurones provide the main output of the nuclei. Small GABAergic neurones innervate the contralateral inferior olive. Both GABAergic and glycinergic interneurones have been identified (Uusisaari and Knöpfel 2011); as far as we know, all cell types receive an inhibitory input from Purkinje cells and an excitatory input from mossy and climbing fibre collaterals.

CEREBELLAR CIRCUITRY

The main circuitry of the cerebellum was described by Ramón y Cajal in the late nineteenth century and published in his *Histologie du système nerveux* (1906). It involves two extracerebellar afferent systems (climbing fibres and mossy fibres), intrinsic cortical neurones, including Purkinje, granule, stellate and basket cells, and neurones in the cerebellar nuclei (see Figs 22.7–22.9). The widely diverging mossy fibre-parallel fibre system terminates on Purkinje cells (the only output neurones of the cortex projecting to the cerebellar and vestibular nuclei); a climbing fibre terminates on the proximal, smooth Purkinje cell dendrites; Golgi cells provide backward inhibition to granule cells; and stellate and basket cells provide forward inhibition to Purkinje cells.

Climbing fibres and most mossy fibres are excitatory and use glutamate as their neurotransmitter. All climbing fibres take their origin from the contralateral inferior olivary nucleus in the medulla oblongata. In the cerebellum, they split into several branches, each branch providing a climbing fibre to a single Purkinje cell. The branches of a single olivocerebellar fibre innervate one or more sagittally orientated strips

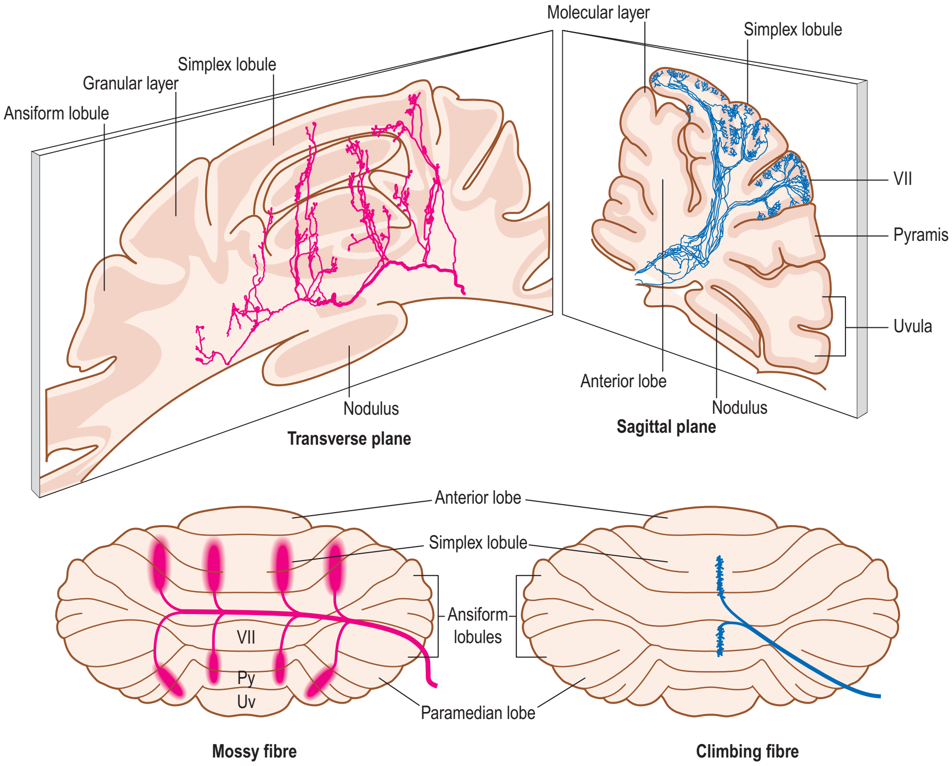

Fig. 22.11 The orientation and branching pattern of mossy and climbing fibres. **Left-hand panels**: mossy fibres are orientated transversely. They distribute bilaterally and emit collaterals at specific, symmetrical locations. These collaterals terminate as sagittally orientated aggregates of mossy fibre terminals. **Right-hand panels**: olivocerebellar fibres branch in the sagittal plane. Each branch provides a Purkinje cell with a single climbing fibre. These climbing fibres form narrow, longitudinally orientated strips that may correspond to the microzones; strips of Purkinje cells that share the same climbing fibre receptive fields. Abbreviations: Py, pyramis; Uv, uvula. (Reproduced with permission from Nieuwenhuys, R, Voogd J, van Huijzen 2008 The Human Nervous System. 4th Ed Springer Verlag.)

of Purkinje cells (**Fig. 22.11**). These strips probably correspond to 'microzones' consisting of a narrow strip of Purkinje cells innervated by climbing fibres sharing the same receptive field. Microzones, with their Purkinje cells, are considered to be the basic structural and functional unit of the cerebellar cortex (Andersson and Oscarsson 1978).

Mossy fibres take their origin from multiple sources in the spinal cord and brainstem. Their myelinated axons terminate on the claw-like dendrites of the granule cells and on Golgi cells. The granule cells give rise to an ascending axon that splits in the molecular layer into two parallel fibres that run for some distance in the direction of the long axis of the folia. Parallel fibres terminate on the spines of the spiny branches of the Purkinje cell dendritic tree and the dendrites of interneurones that they meet along their course (see Figs 22.7, 22.9). The length of the parallel fibres in the human cerebellar cortex is not known but the two branches probably do not exceed 10 mm. Like the climbing fibres, mossy fibres branch profusely in the cerebellar white matter (see Fig. 22.11). The parent fibres enter the cerebellum laterally and run a transverse course to decussate in the cerebellar commissure. During their course, they emit thin collaterals that enter the white matter of the lobules and terminate in multiple, longitudinally orientated and symmetrically distributed aggregates of mossy fibre terminals in the granular layer (Wu et al 1999).

The climbing fibre microzones and the subjacent mossy fibre aggregates have been found to share the same peripheral receptive fields in regions of the cerebellum receiving somatosensory information from the periphery (Ekerot and Larson 1980, Ekerot and Jörntell 2003). A similar topographical relationship between microzones and mossy fibre terminal aggregates exists in other parts of the cerebellar cortex; their common denominator remains unknown (Pijpers et al 2006). The significance of such a topographical relationship is difficult to understand because the parallel fibres would disperse a localized mossy fibre input over a wide, mediolateral region of the molecular layer. Different hypotheses to explain this topographical relationship have been proposed, some of the more recent ones involving the interneurones of the cerebellar cortex, but the matter remains undecided (Ekerot and Jörntell 2003, Barmack and Yakhnitsa 2011).

THE MODULAR ORGANIZATION OF THE CEREBELLUM AND THE CORTICONUCLEAR AND OLIVOCEREBELLAR PROJECTIONS

The output of the cerebellum is organized as a series of parallel, sagittal modules (Voogd and Bigaré 1980, Voogd and Ruigrok 2012). Each module consists of one or more longitudinal Purkinje cell zones that project to one of the cerebellar or vestibular nuclei (**Fig. 22.12A**). Some of these Purkinje cell zones are restricted to certain lobules; others span the entire rostrocaudal length of the cerebellum. Climbing fibres from a subdivision of the contralateral inferior olive terminate on the Purkinje cells of a particular zone and also send a collateral innervation to the corresponding deep cerebellar nucleus. This collateral innervation is reciprocated by the, mainly crossed, nucleo-olivary pathway that originates from the small GABAergic neurones of the cerebellar nuclei. Modules can be visualized because their Purkinje cell axons and their climbing fibre afferents collect in compartments in the cerebellar white matter. The borders between these compartments, i.e. between the modules, become visible when stained for acetylcholine esterase (AChE) (**Fig. 22.12B**). The modular organization of the cerebellum has been studied in most detail in rodents and carnivores, and has been confirmed in non-human primates. For the human cerebellum, evidence for its presence is mainly embryological.

The modular organization of the cerebellum appears very early during its development, long before the emergence of any of its transverse fissures. Purkinje cells, born in the ventricular matrix of the cerebellar anlage, migrate to the meningeal surface, where they form a series of mediolaterally arranged clusters (Korneliussen 1968, Kappel 1981). During the later increase of the cerebellar surface in the rostrocaudal dimension, reflecting the proliferation of millions of granule cells in the transient external matrix (the external granular layer) (see Fig. 22.15), the Purkinje cell clusters increase in length and thus are transformed into Purkinje cell zones. The Purkinje cells become located in a monolayer and the original borders between the clusters are no longer visible. This mode of development has been studied in different species and can also be recognized in the human fetal cerebellum. Purkinje cell clustering in the human does not differ from that in other species, with the exception of the immense size of the most lateral cluster, which is clearly related to the anlage of the dentate nucleus (**Fig. 22.13**). This cluster develops into the D2 zone, the most lateral Purkinje cell zone, responsible for the large size of the human cerebellar hemisphere.

Eight or nine of the modules can be recognized in the cerebellum of subhuman primates and lower mammals (see Fig. 22.14A). Purkinje cell zones differ in their climbing fibre afferents and their cerebellar or vestibular target nucleus. Moreover, Purkinje cells of the different zones differ in their immunohistochemical properties (Voogd and Ruigrok 2012). A Purkinje cell-specific antibody, known as 'zebrin 2', is distributed in a pattern of zebrin-positive and zebrin-negative Purkinje cell zones (**Fig. 22.14D–E**). This pattern has been shown to be congruent with the olivocerebellar and corticonuclear projection zones (Voogd et al 2003, Sugihara and Shinoda 2004). Many substances, such as the enzymes aldolase C, 5′ nucleotidase, protein kinase C and the metabotropic glutamate transporter 1A, co-localize with zebrin 2. Neurotransmission in different Purkinje cell zones may therefore differ: zebrin-positive Purkinje cells fire at a slower rate than the zebrin-negative cells (Zhou et al 2014).

In the following, the olivocerebellar climbing fibre and the efferent corticonuclear projections of the Purkinje cell zones will be described.

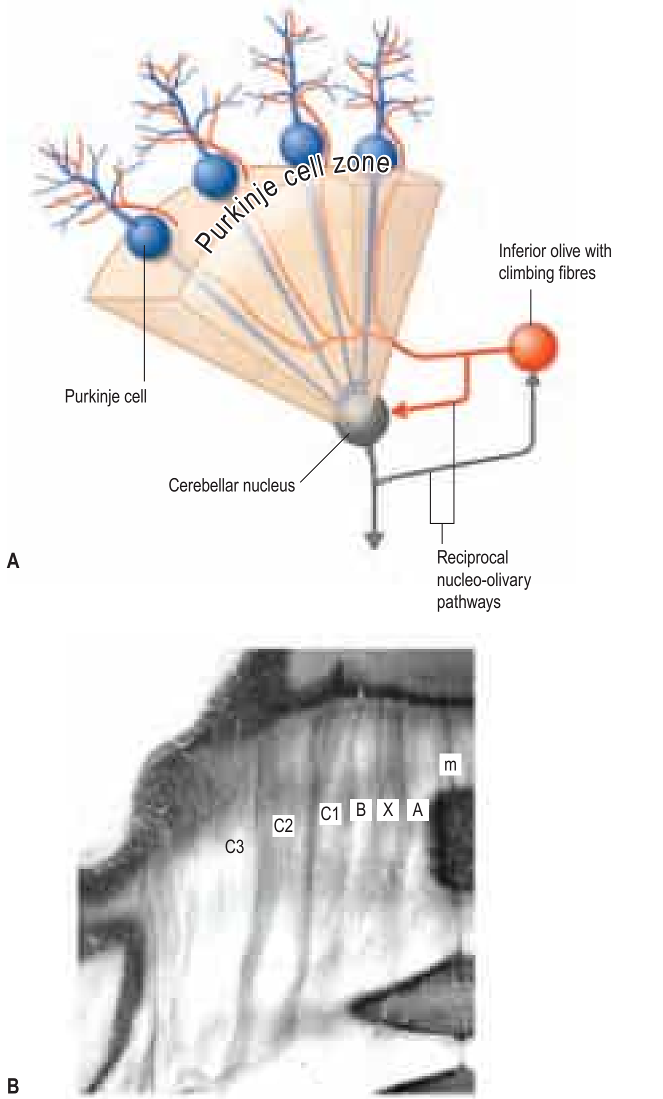

Fig. 22.12 **A**, A cerebellar module. Purkinje cell axons and climbing fibres are located in a white matter compartment, shown as a transparent structure in this diagram. **B**, An acetylcholinesterase-stained section through the anterior lobe: macaque monkey. The borders of the white matter compartments of the modules A–C are heavily stained. Abbreviations: m, midline.

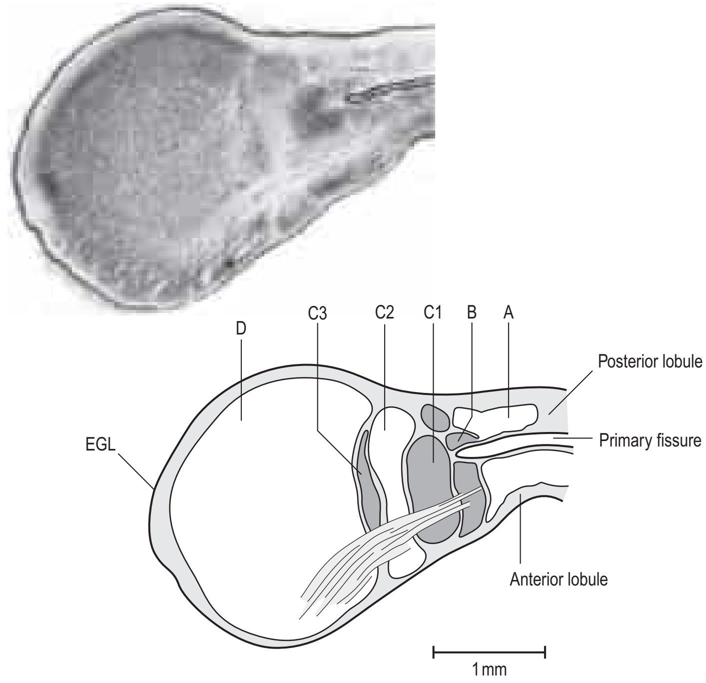

Fig. 22.13 A transverse section through the cerebellum of a 65 mm human fetus, showing the Purkinje cell clusters that will develop into the A, B, C1–C3 and D Purkinje cell zones. Note the large size of the D cluster. Abbreviations: EGL, external granular layer. (From the Schenk collection of the Dept. of Pathology of the Erasmus Medical Center Rotterdam.)

Data are from experimental studies in the cat, the rat and subhuman primates (reviewed in Voogd and Ruigrok (2012)). The subdivision of the inferior olive, the sole source of the climbing fibres, is summarized in **Figure 22.15**.

The A zone is located next to the midline and extends over the entire vermis (see Fig. 22.14A). It is composed of several zebrin-positive and zebrin-negative subzones that may be present over limited segments of its extent. It projects to the fastigial and vestibular nuclei, and receives its climbing fibres from the caudal medial accessory olive. Whereas the A zone extends over the entire vermis, the X and B zones are restricted to the vermis of the anterior lobe, the simplex lobule (VI) and lobule VIII (the pyramis). The narrow X zone separates the A zone from the B zone, which occupies the lateral vermis of these lobules. The X zone projects to the interstitial cell groups, located between the fastigial and posterior interposed nuclei and receives climbing fibres from the intermediate region of the medial accessory olive. The B zone projects to Deiters' lateral vestibular nucleus and is innervated by climbing fibres from the caudal part of the dorsal accessory olive. The dorsal accessory olive, the B zone and the lateral vestibular nucleus are somatotopically organized. In the B zone, the hindlimb is represented laterally and the forelimb is represented medially (Andersson and Oscarsson 1978). In rodents, Purkinje cells of the X and B zones are zebrin-negative.

The hemisphere is composed of the C1–C3 and the D1, Y and D2 zones. Like the X and B zones, C1, C3 and the Y are restricted to the anterior lobe, the simplex lobule (HVI) and the paramedian lobule (HVIIB – the gracile lobule, and HVIII – the biventral lobule). Moreover, the Purkinje cells of these zones are zebrin-negative and thus appear as blank spaces in suitably immunostained histological sections of the anterior and posterior cerebellum (see Fig. 22.14E). The C1, C3 and Y zones project to the anterior interposed nucleus and receive their climbing fibre input from the rostral dorsal accessory olive (DAOr) (**Fig. 22.16**). This subnucleus receives an input from peripheral receptors through dorsal column and trigeminal pathways. The climbing fibre projections of the rostral dorsal accessory olive to the C1, C3 and Y zones and the anterior interposed nucleus are somatotopically organized. In each of the zones, the hindlimb is represented rostrally in the anterior cerebellum and caudally in the posterior lobe; the forelimb and face occupy more central areas (Ekerot and Larson 1979) (see Fig. 22.16). This rostrocaudal distribution clearly differs from the mediolateral somatotopy in the vermal B zone. The somatotopical localization is an extremely detailed one, repeated in each of the zones. The C1, C3 and Y zones connect with motor centres in the brainstem and the cerebral cortex. The hemisphere of the anterior lobe and the simplex lobule, and the paramedian lobule (HVIIB – the gracile lobule, and HVIII – the biventral lobule) are considered as the motor regions of the cerebellum.

The C2, D1 and D2 zones extend beyond the anterior and posterior motor regions, where they interdigitate with the C1, C3 and Y zones, over most of the rostrocaudal length of the cerebellum. In rodents, these zones are zebrin-positive. The C2 zone projects to the posterior interposed nucleus and receives its climbing fibre input from the rostral medial accessory olive (see Fig. 22.18). A somatotopical organization is lacking in the C2 zone. The D1 and D2 zones project to the caudal and rostral dentate and receive their climbing fibres from the ventral and dorsal laminae of the principal olive, respectively. The main connections of the C2 and the D zones are with the cerebral cortex. The sections of the C2 and D2 zones located in the anterior and posterior motor regions of the cerebellar hemisphere are connected with motor, premotor and parietal cortical areas; these sections of the D2 zone are somatotopically organized. Sections of the C2 and the D zones located in the ansiform lobule (HVII) and the paraflocculus (the tonsil, HIX) subserve visuomotor and non-motor functions.

The modular organization of the vestibulocerebellum is fairly complex; multiple Purkinje cell zones, innervated by climbing fibres from subnuclei in the inferior olive, transmit optokinetic and vestibular information.

Each lobule of the cerebellum contains a particular set of Purkinje cell zones. Apart from the parallel fibres, which may cross several Purkinje cell zones or microzones in their course through the molecular layer, there is no cross-talk between the modules. Parallel fibres are, therefore, a key element in the integrative function of the cerebellum. The relative independence of the cerebellar modules is an important

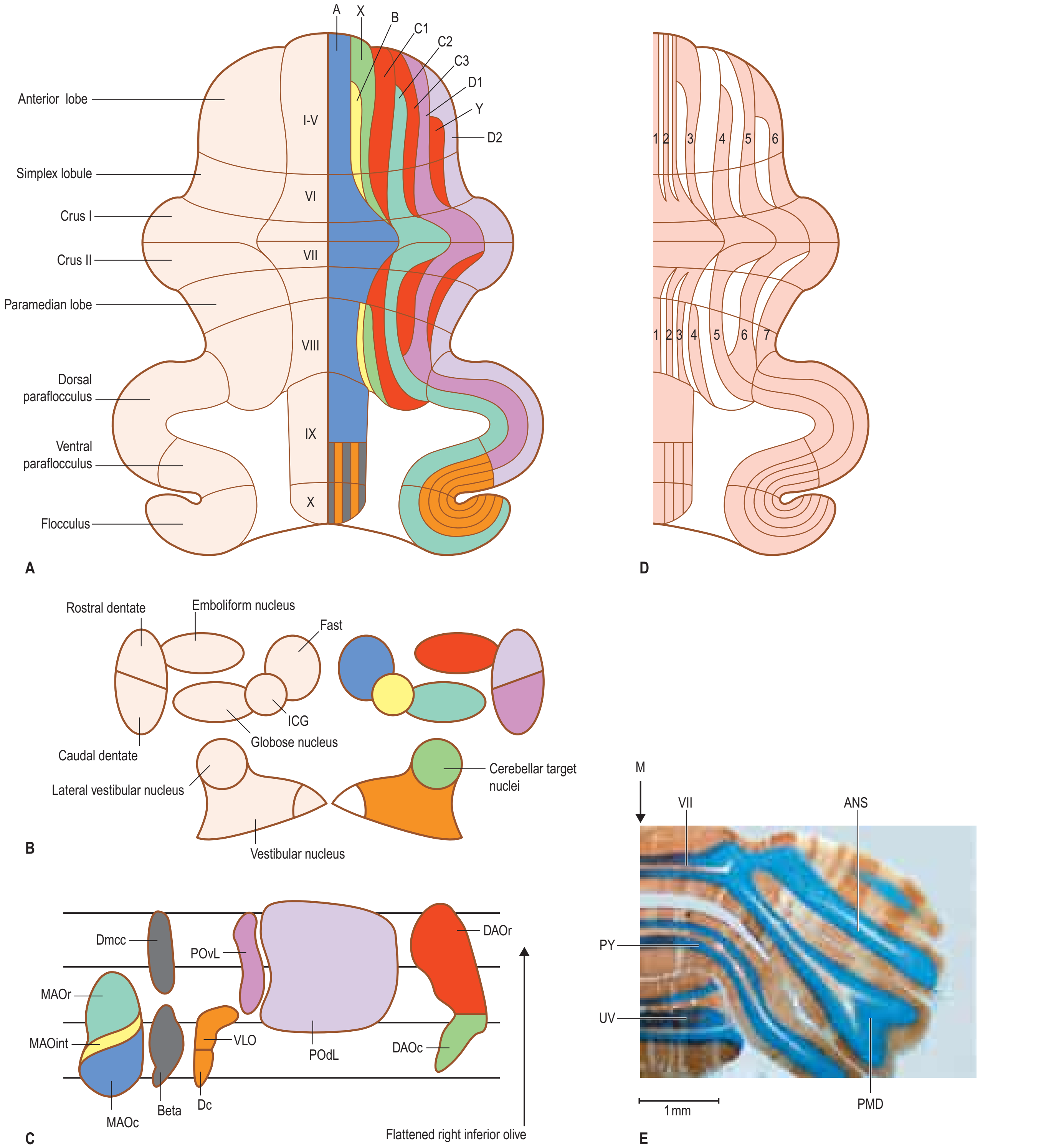

Fig. 22.14 The connections of the Purkinje cell zones of the mammalian cerebellum. **A**, The flattened cerebellar cortex. **B**, Target nuclei of Purkinje cells. **C**, Sources of climbing fibres associated with Purkinje cells, shown in the flattened contralateral inferior olive (see Fig. 22.15) and indicated in the same colour. **D**, Zebrin-positive and zebrin-negative Purkinje cell bands. The zebrin-positive bands are numbered from 1 to 7. A comparison with panel A shows that the A zone is a composite of zebrin-positive and zebrin-negative subzones; the X, B, C1, C3 and Y zones consist of zebrin-negative Purkinje cells. **E**, The zebrin-positive and zebrin-negative bands of the cerebellum of a rat. Abbreviations: A–D2, Purkinje cell zones A–D2; ANS, ansiform lobule; Beta, group beta; DAOc/r, caudal/rostral dorsal accessory olive; Dc, dorsal cap; Dmcc, dorsomedial cell column; Fast, fastigial nucleus; ICG, interstitial cell groups; M, midline, MAOc/int/r, caudal/intermediate/rostral medial accessory olive; PMD, paramedian lobule; POdL, dorsal lamina of the principal olive; POvL, ventral lamina of the inferior olive; PY, pyramis (lobule VIII); UV, uvula (lobule IX); VII, vermal lobule VII; VLO, ventrolateral outgrowth.

difference between the cerebellum and the cerebral cortex, where different functional areas are intimately interconnected.

CONNECTIONS OF THE CEREBELLAR NUCLEI: RECIPROCAL ORGANIZATION OF THE CORTICO-OLIVARY SYSTEM

The connections of the cerebellar nuclei with the brainstem, the thalamus and the spinal cord determine the sphere of influence of the cerebellar modules. The A, X and B zones of the vermis project to the fastigial nucleus, the interstitial cell groups and the lateral vestibular nucleus, respectively (see Fig. 22.14). In all mammals, the fastigial nucleus gives rise to the uncinate tract, which decussates in the cerebellar commissure, hooks around the brachium conjunctivum, and is distributed to the vestibular nuclei and the medullary and pontine reticular formation. A branch of the uncinate tract ascends to the ipsilateral midbrain and thalamus. Projections to the cerebral cortex are bilateral because the crossed ascending fibres of the uncinate fasciculus subsequently recross in the thalamus. Their projection to the cerebral cortex is incompletely known. The uncrossed, direct fastigiobulbar tract passes along the lateral margin of the fourth ventricle. It is distributed

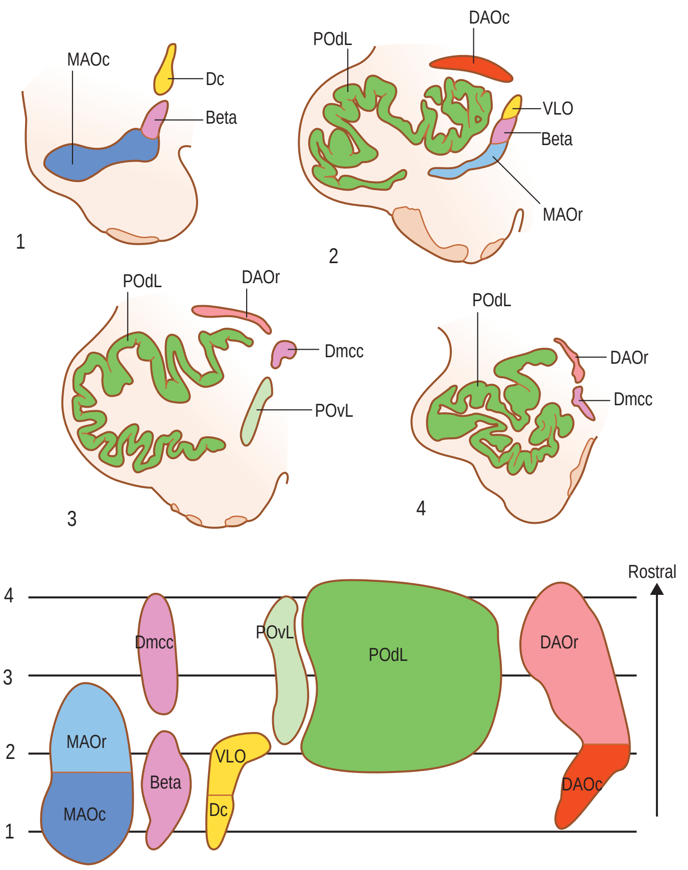

Fig. 22.15 Transverse sections through the human inferior olive, section 1 being the most rostral. Lower panel: the flattened inferior olive showing the levels of sections 1–4 in the upper panel. Note the large size of the convoluted dorsal lamina of the principal olive (POdL), and the small ventral lamina (POvL). Other abbreviations: Beta, group beta; DAOc/r, caudal/rostral dorsal accessory olive; Dc, dorsal cap; Dmcc, dorsomedial cell column; MAOc/r, caudal/rostral medial accessory olive; VLO, ventrolateral outgrowth.

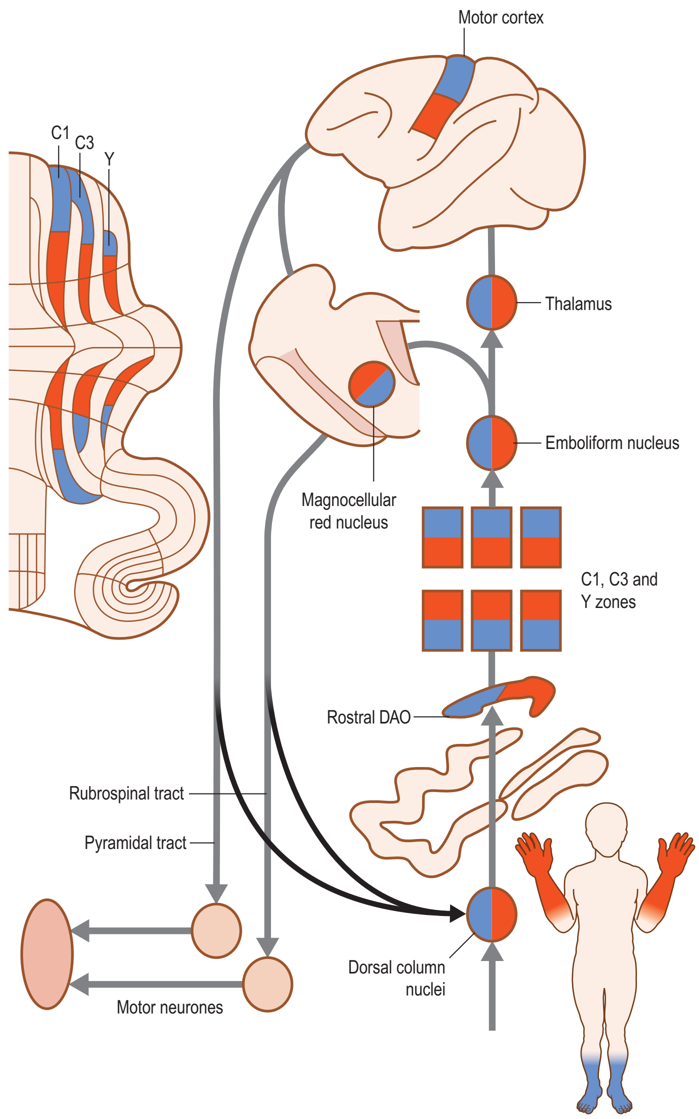

Fig. 22.16 The connections of the emboliform (anterior interposed) nucleus. The entire system is somatotopically organized: this organization is more detailed than indicated in the diagram. Abbreviations: rostral DAO, rostral dorsal accessory olive.

to the vestibular nuclei and the reticular formation in a symmetrical manner that mirrors that of the uncinate tract (Batton et al 1977). The direct fastigiobulbar tract is an inhibitory, glycinergic system (Bagnall et al 2009). Small GABAergic neurones give rise to a nucleo-olivary pathway terminating in the contralateral caudal medial accessory olive.

The caudal pole of the fastigial nucleus receives its Purkinje cell afferents from lobule VII (folium and tuber vermis). This lobule is also known as the visual vermis because it is involved in the long-term adaptation of saccades and, possibly, in other eye movements. The projections of the oculomotor region of the fastigial nucleus (**Fig. 22.17**) are completely crossed. They terminate in the pontine paramedian reticular formation (the horizontal gaze centre), the superior colliculus, the rostral interstitial nucleus of the medial longitudinal fasciculus (the vertical gaze centre) and in the thalamic nuclei that may include the frontal and parietal eye fields as their targets (Noda et al 1990). The fastigial nucleus influences visceromotor systems via projections of the vestibular nuclei and connections with the catecholaminergic nuclei of the brainstem and the hypothalamus (Zhu et al 2006).

The projections of the interstitial cell groups located between the fastigial and posterior interposed nuclei, the target nucleus of the X module, have not been studied in primates. In lower mammals, these neurones provide collaterals to the superior colliculus, thalamus and spinal cord (Bentivoglio and Kuypers 1982).

The lateral vestibular nucleus (Deiters' nucleus) is the target nucleus of the lateral vermal B zone. This nucleus might better be considered as one of the cerebellar nuclei. It does not receive a primary input from the labyrinth and, contrary to the other vestibular nuclei, receives a collateral innervation from the climbing fibres innervating the B zone. It gives rise to the lateral vestibulospinal tract. Its nucleo-olivary pathway targets the caudal dorsal accessory olive.

The zones of the cerebellar vermis are in a position to affect neurotransmission in the vestibulospinal and reticulospinal systems, bilaterally controlling postural and vestibular reflexes of the axial and proximal musculature, and in the oculomotor centres in the brainstem. The skeletomotor and oculomotor functions are located in specific segments of the vermis: skeletomotor functions in the anterior vermis and posterior lobule VIII (pyramis) (the X and B zones are restricted to these lobules), and oculomotor functions in lobule VII. Caudalmost, lobule X (nodulus) belongs to the vestibulocerebellum and is considered below. Other functions, such as vegetative regulation, are subserved by the vermis but have not been studied in detail.

The anterior interposed (emboliform) nucleus is the target of the C1, C3 and Y zones. The detailed somatotopical organization of these Purkinje cell zones is maintained in the anterior interposed nucleus, where Purkinje cells of different zones, but with the same climbing fibre input from a particular region of the body, project to a common set of neurones (see Fig. 22.16). Ascending axons from the anterior interposed nucleus enter the brachium conjunctivum. This tract decussates at the border of the pons and the mesencephalon. The ascending branch enters and surrounds the magnocellular red nucleus and proceeds to the thalamus, from where the anterior interposed nucleus is connected with the contralateral primary motor cortex. The descending branch of the brachium conjunctivum terminates in the nucleus reticularis tegmenti pontis (reticular tegmental nucleus of the pons). The entire system, including the magnocellular red nucleus and the primary motor cortex and their efferent tracts, is somatotopically organized. A nucleo-olivary pathway from the anterior interposed nucleus terminates in the rostral dorsal accessory olive.

The motor cortex and the magnocellular red nucleus give rise to the two main descending motor systems: the corticospinal (pyramidal) tract and the rubrospinal tract. Both of these tracts cross the midline, the former at the bulbospinal junction and the latter at its level of origin

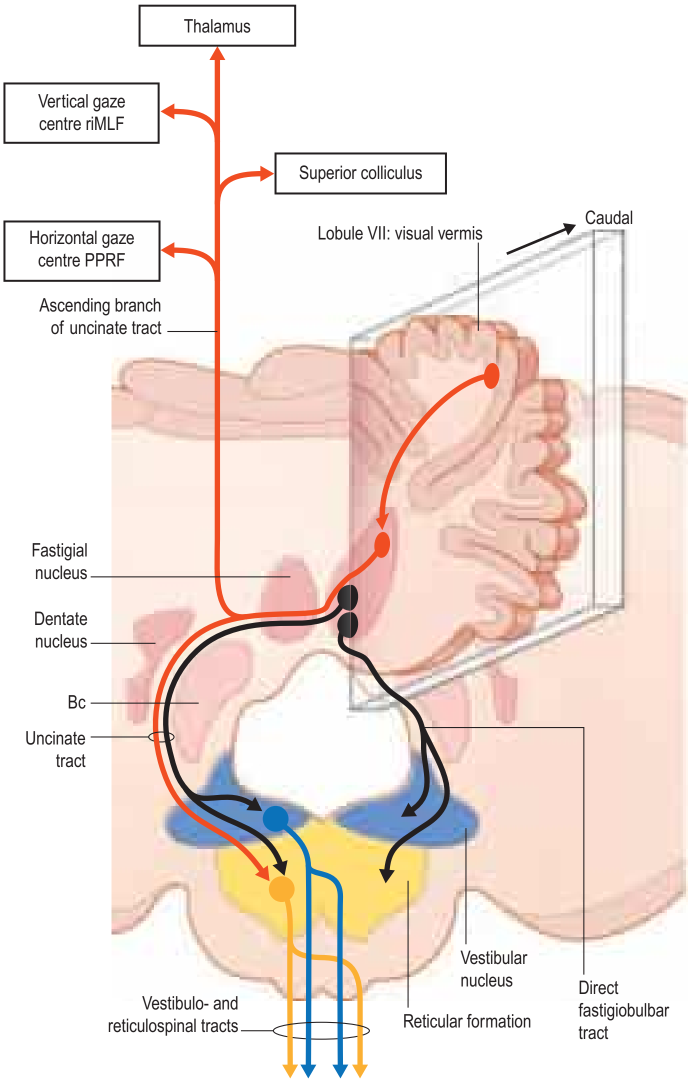

Fig. 22.17 A transverse section through the cerebellum and medulla oblongata, showing the symmetrical distribution of the crossed and uncrossed connections of the fastigial nucleus. The inset depicts a sagittal section, showing the connections of the visual vermis (lobule VII) with the caudal pole of the fastigial nucleus and its efferent pathways in red. Abbreviations: Bc, brachium conjunctivum; PPRF, paramedian pontine reticular formation; riMLF, rostral interstitial nucleus of the medial longitudinal fasciculus.

in the midbrain. The corticospinal tract provides the magnocellular red nucleus with a collateral innervation. Both tracts influence distal movements of the limbs. During primate evolution, the corticospinal system increases in prominence at the cost of the rubrospinal system, which comes to occupy a subsidiary position in the human brain.

Climbing fibres innervating the C1, C3 and Y zones and the anterior interposed nucleus take their origin from the rostral dorsal accessory olive, which receives a somatotopically organized cutaneous input, mainly through the dorsal column and trigeminal nuclei, and contains a refined cutaneous map of the entire contralateral body surface (Gellman et al 1983). The corticospinal and rubrospinal tracts provide the dorsal column nuclei with a collateral innervation.

It should be emphasized that the concept of the cerebellum as a motor system is closely allied to the circuitry of the C1, C3 and Y zones, and to the anterior interposed nucleus and its output systems. The double decussation of the brachium conjunctivum and the rubrospinal and corticospinal tracts is responsible for the clinical observation that lesions of the cerebellum affect the ipsilateral half of the body. For most of the other modules with predominantly cerebral cortical connections, the functional relations are much less clear.

The connections of the posterior interposed (globose) and dentate nuclei are arranged according to the same plan. They ascend and decussate in the brachium conjunctivum, and terminate in a group of nuclei at the mesodiencephalic junction that includes the parvocellular red nucleus and the nucleus of Darkschewitsch in the central grey, and in the thalamic nuclei that project to motor, premotor, prefrontal and posterior parietal cortical areas and the frontal and parietal eye fields (Fig. 22.18A). The nuclei at the mesodiencephalic junction give rise to the ipsilaterally descending tegmental tracts that terminate in the inferior olive, forming reciprocally organized loops; the function of these prominent recurrent loops has never been studied.

The posterior interposed (globose) nucleus projects to the nucleus of Darkschewitsch and, via the thalamus, to most, if not all, cortical areas (Fig. 22.18B). Reciprocal connections of the cerebral cortex to the nucleus of Darkschewitsch have been reported for most cortical areas. The nucleus of Darkschewitsch gives rise to a recurrent climbing fibre loop to the C2 zone that consists of the medial tegmental tract and the rostral medial accessory olive. Motor and visual divisions can be distinguished in this system. The segments of the C2 zone located in the anterior and posterior motor regions of the cerebellum and the rostromedial posterior interposed nucleus receive input from the motor cortex. Visual and prefrontal input dominates in segments located in the ansiform lobule (HVII), the paraflocculus (HIX) and the flocculus (HX). The nucleo-olivary pathway from the posterior interposed nucleus innervates the contralateral rostral medial accessory olive.

The rostral and caudal dentate nucleus give rise to different pathways. Neurones of the caudal pole of the dentate nucleus are known to be activated by eye movements (van Kan et al 1993). The caudal dentate projects to a dorsomedial subnucleus of the parvocellular red nucleus, located medial to the fasciculus retroflexus (Fig. 22.18C). Its thalamocortical projections include the frontal and parietal eye fields, which are reciprocally connected with the dorsomedial subnucleus. The latter projects to the ventral lamina of the principal olive, which innervates the D1 zone. Although fairly prominent in lower mammals, the ventral lamina of the principal olive is represented by the narrow medial lamina of the human olive (see Fig. 22.15). This module, presumably, is much reduced in the human cerebellum. Crossed nucleo-olivary pathways from the rostral and caudal dentate terminate in the dorsal and ventral laminae of the principal olivary nucleus, respectively.

The rostral dentate includes the major part of the dentate nucleus. In monkeys, it has been divided into rostromedial motor and ventrocaudal non-motor portions (Strick et al 2009) (Fig. 22.18E). The motor division is somatotopically organized, with the hindlimb represented rostrally and the face more caudally; it receives projections from motor regions of the cerebellum. The caudal non-motor portion receives its corticonuclear projections from the ansiform lobule (HVII) and the paraflocculus (the tonsil, HIX). A similar subdivision of the dentate has been proposed in humans (Küper et al 2012); it seems likely that these divisions correspond with the rostromedial microgyric and ventrocaudal macrogyric regions of the human dentate (see Fig. 22.10).

The rostral dentate projects to the major, ventrolateral, portion of the parvocellular red nucleus. Its thalamocortical projections target the motor, premotor and posterior parietal cortices (Fig. 22.18D–E). Projections from the caudal dentate include the dorsal prefrontal cortex. Reciprocal connections between these cortical areas and the parvocellular red nucleus have been documented, mainly for the motor and premotor areas. These projections are somatotopically organized; they occupy the lateral portion of the parvocellular red nucleus. Prefrontal projections are located more medially. The parvocellular red nucleus connects with the dorsal lamina of the principal olivary nucleus through the central tegmental tract. Motor input is transmitted by the dorsal lamina to segments of the D2 zone located in the motor regions of the cerebellum; non-motor input is transmitted to the ansiform lobule (HVII) and the tonsil (HIX).

In humans, the D2 zone accounts for most of the cerebellar hemisphere. This is exemplified by its development (see Fig. 22.13) and by the size of the different components of its circuitry. In Figure 22.19, the first ever published lithograph of a section through the pontine tegmentum (Stilling 1846), the central tegmental tract can be recognized immediately as one of the largest fibre systems in the brainstem. Several explanations have been offered for the size of the dentate and its connections. They include the complexity of the cortical motor system, which is a major target of the dentate nucleus. (Multiple, interconnected premotor and posterior parietal areas involved in the preparation of movement converge on the primary motor cortex; the precise contribution of the cerebellum to these processes is not known.) Other possible explanations are the increased connectivity of the dentate with the prefrontal cortex subserving its non-motor functions (Stoodley and Schmahmann 2009), and an increase in the dentate-parietal projection, given that non-motor functions also involve the parietal cortex.

During evolution, the shapes of the dentate and the principal olivary nucleus change from compact nuclei to intricately folded sheets. This may indicate the presence of a detailed topical localization in the corticonuclear and climbing fibre afferent connections in the D2 zone, but almost nothing is known about its intrinsic organization.

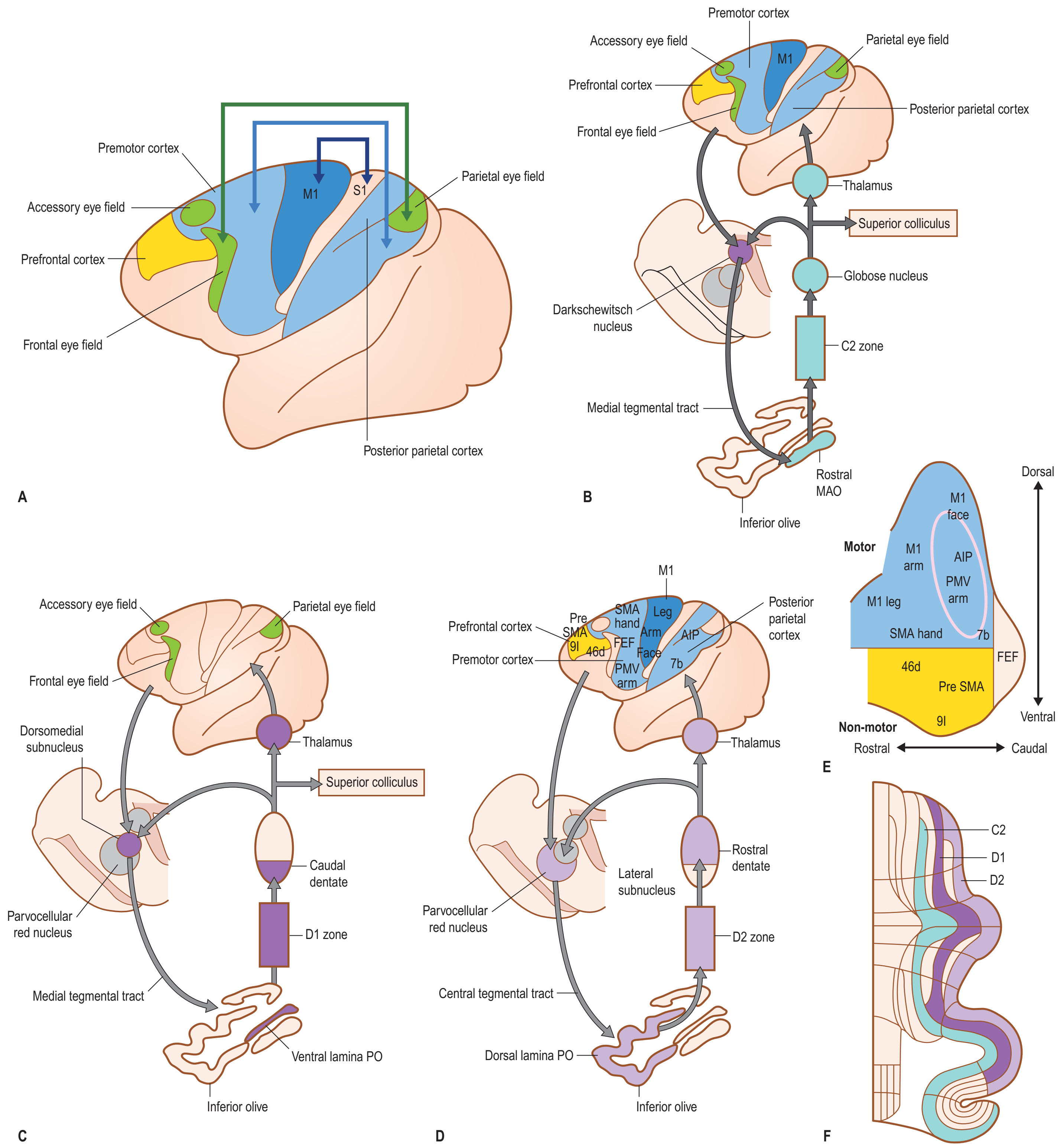

Fig. 22.18 **A**, Cortical areas targeted by the cerebellothalamic pathways of the posterior interposed and dentate nuclei. The primary motor area (M1) with the primary sensory area (S1), the premotor cortex with the posterior parietal areas and the frontal and parietal eye fields constitute interconnected networks. **B**, The connections of the globose (posterior interposed) nucleus. **C**, The connections of the caudal pole of the dentate nucleus. **D**, The connections of the rostral dentate nucleus. **E**, The subdivision of the rostral dentate into rostral motor and caudal non-motor divisions, showing the location of neurones retrogradely labelled from injection sites indicated in the diagram of the cerebral cortex in D. **F**, The flattened cerebellar cortex showing localization of the C2, D1 and D2 zones. Abbreviations: AIP, anterior intraparietal area; FEF, frontal eye field; PMV, ventral premotor cortex; PO, principal olive; rostral MAO, rostral medial accessory olive; SMA, supplementary motor area; 7b, 46d, 9l, cortical areas. (D, Modified with permission from Strick PL, Dum RP, Fiez JA 2009 Cerebellum and nonmotor function. Annu Rev Neurosci 32:413–434.)

AFFERENT MOSSY FIBRE CONNECTIONS OF THE CEREBELLUM

Mossy fibre systems take their origin from multiple sites in the spinal cord and the brainstem. The pontocerebellar pathway is the major mossy fibre system in primates. Although mossy fibre systems have rarely been traced with experimental methods in primates, fMRI has provided information on their organization in the human cerebellum.

Mossy fibre systems share several common features. Individual mossy fibres distribute bilaterally and give off collaterals at specific mediolateral positions that terminate in longitudinal aggregates of mossy fibre rosettes (see Fig. 22.11). Entire mossy fibre systems terminate as multiple, bilaterally distributed bands of mossy fibre terminals (Fig. 22.20A). These bands are not continuous, but are often restricted to either the apices or the bases of the folia. Exteroceptive components of mossy fibre systems terminate superficially, whereas proprioceptive systems terminate in the bases of the folia (Ekerot and Larson 1972)

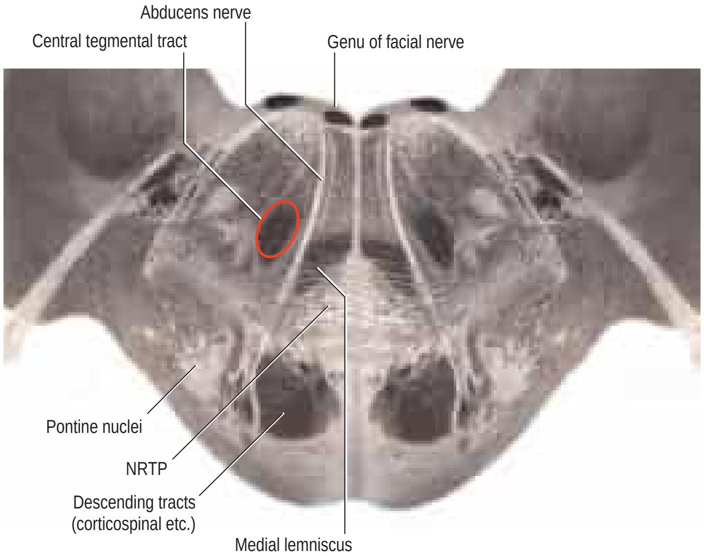

Fig. 22.19 A lithograph of a transverse section through the pons, showing the localization of the central tegmental tract in the pontine tegmentum. Abbreviation: NRTP, nucleus reticularis tegmenti pontis. (Reproduced from Stilling B 1846 Untersuchungen über den Bau und die Verrichtungen des Gehirns. I. Über den Bau des Hirnknotens oder der Varolischen Brücke. Jena, Druck und Verlag von Friedrich Make.)

(Fig. 22.20B). The mossy fibre aggregates are not as distinct as the climbing fibre zones and often merge in the bases of the fissures. Mossy fibre aggregates of different systems interdigitate or overlap; precise information is lacking.

The termination of the spinocerebellar, reticulocerebellar, cuneocerebellar and trigeminocerebellar tracts is restricted to the anterior and posterior motor regions of the cerebellum, i.e. to the anterior lobe, the simplex lobule (VI and HVI), lobule VIII and the paramedian lobule (gracile HVIIB and biventral HVIII lobules). These lobules also receive primary and secondary vestibulocerebellar inputs and pontocerebellar mossy fibres relaying information from cortical motor areas. Many of these mossy fibre systems terminate according to a somatotopical pattern (Fig. 22.20C). A very similar somatotopical organization occurs in the C1, C3 and Y climbing fibre zones that are restricted to the hemisphere of the same lobules (see Fig. 22.16).

Spinocerebellar, trigeminocerebellar, reticulocerebellar and vestibulocerebellar fibres

The spinal cord is connected to the cerebellum through the spinocerebellar and cuneocerebellar tracts, and through indirect mossy fibre pathways relayed by the lateral reticular nucleus in the medulla oblongata. These pathways are all excitatory in nature. Some of them give collaterals to the cerebellar nuclei before ending on cortical granule cells.

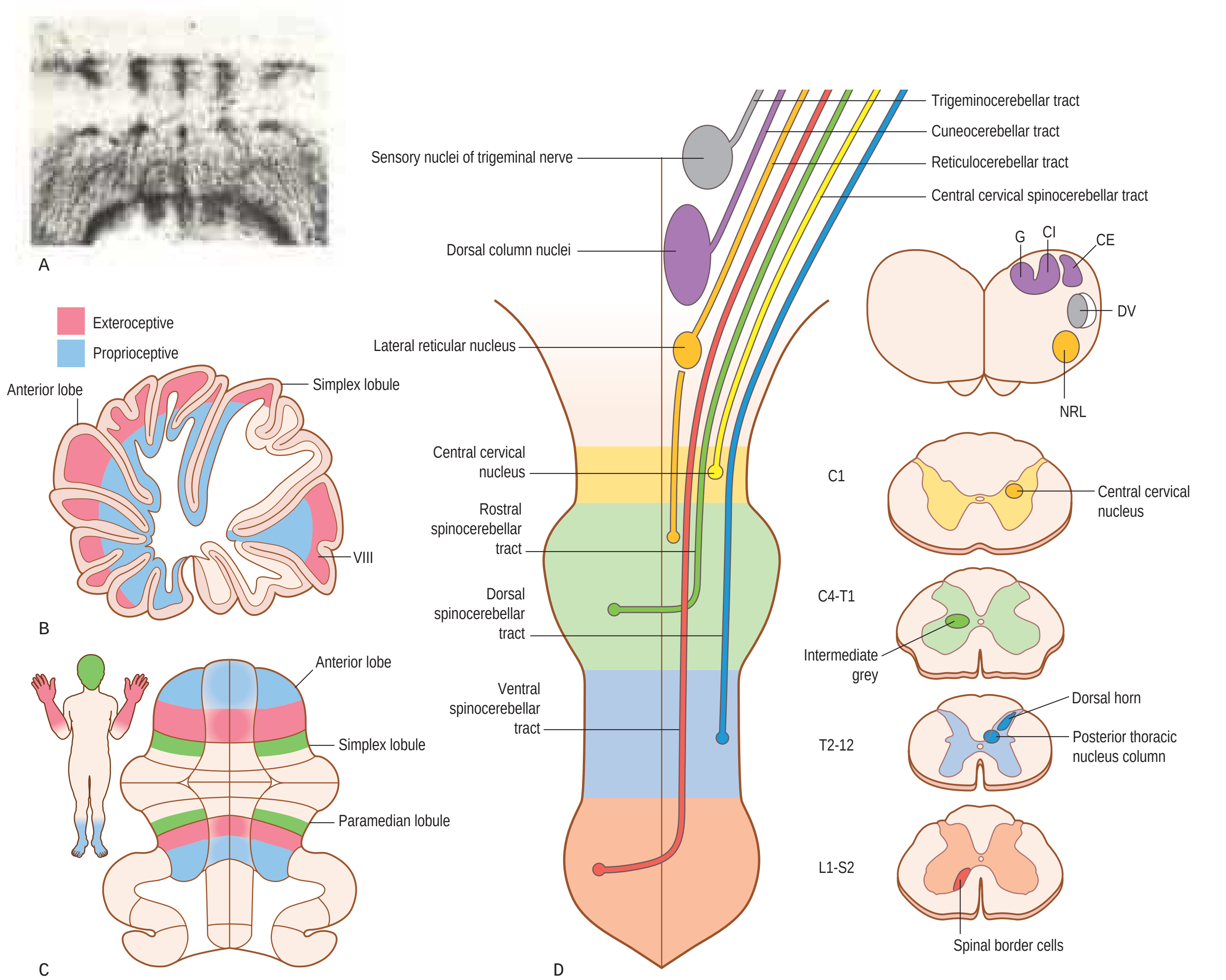

Fig. 22.20 A, The termination of spinocerebellar fibres as multiple sagittal bands in the anterior lobe of *Tupaia glis*. **B**, A sagittal section through the cerebellum showing the termination of exteroceptive mossy fibre systems in the apices of the lobules of the anterior lobe, the simplex lobule and lobule VIII (pyramis), and of proprioceptive systems in the bases of the fissures. **C**, The somatotopic organization of the termination of the exteroceptive components of the spinocerebellar, cuneocerebellar and trigeminocerebellar tracts in the hemisphere of the anterior lobe, the simplex lobule and the paramedian (biventral) lobule. **D**, The origin of the spinocerebellar, cuneocerebellar and reticulocerebellar tracts. Abbreviations: CE, external cuneate nucleus; CI, internal cuneate nucleus; DV, nucleus of the spinal tract of the trigeminal nerve; G, gracile nucleus; NRL, lateral reticular nucleus.

The dorsal spinocerebellar tract transmits information from the ipsilateral lower limb (Ch. 20). It contains proprioceptive fibres that arise from neurones in the posterior thoracic nucleus (Clarke's column) in the thoracic and upper lumbar spinal cord, and exteroceptive fibres from the thoracic and lumbar dorsal horns. It enters the cerebellum in the inferior cerebellar peduncle to terminate bilaterally in the vermis and hemisphere of the anterior and posterior lower limb regions.

The cuneocerebellar tract is considered as the upper limb equivalent of the dorsal spinocerebellar tract (Ch. 20). It takes its origin from the dorsal column nuclei, the exteroceptive component from the internal cuneate and gracile nuclei, and the proprioceptive component from the external cuneate nucleus. Both components terminate in the anterior and posterior upper limb regions: the proprioceptive component bilaterally in the bases of the fissures, and the exteroceptive component ipsilaterally in the apices of the lobules of the hemisphere. The exteroceptive component has been shown to terminate in multiple longitudinal zones congruent with the climbing fibre zones of this region; these zones share the same detailed somatotopical organization as the C1, C3 and the Y climbing fibre zones (Ekerot and Larson 1980).

The ventral spinocerebellar tract is a composite pathway. It informs the cerebellum about the state of activity of spinal reflex arcs related to the lower limb and lower trunk. Its fibres originate in the intermediate grey matter and the spinal border cells of the lumbar and sacral segments of the spinal cord, cross near their origin, and ascend close to the surface as far as the lower midbrain before looping around the entrance of the trigeminal nerve to join the superior cerebellar peduncle. Most of these fibres cross again in the cerebellar white matter.

The rostral spinocerebellar tract originates from cell groups of the intermediate zone and horn of the contralateral cervical enlargement. Although considered to be the upper limb and upper trunk equivalent of the ventral spinocerebellar tract, most of its fibres remain ipsilateral throughout their course. They enter the cerebellum through both the superior and the inferior cerebellar peduncles and terminate in the anterior and posterior vermis.

An upper (central) cervical spinocerebellar tract originates from a central cervical nucleus at high cervical levels (C1–C4). The tract terminates bilaterally in the bases of the fissures of the entire cerebellum, lacks a somatotopical organization and conveys labyrinthine information and proprioception from neck muscles (Matsushita and Tanami 1987).

Trigeminocerebellar mossy fibres stem from the ipsilateral principal sensory nucleus and the nucleus of the spinal tract of the trigeminal nerve, and terminate in the hemisphere in the anterior and posterior face regions (simplex lobule – HVI – and gracile lobule – HVIIB).

The distinct somatotopic organization of the anterior and posterior motor regions of the hemispheres is reflected in the termination of the exteroceptive components of the dorsal spinocerebellar, cuneocerebellar and trigeminocerebellar tracts. It is much less distinct for proprioceptive systems, such as the central cervical spinocerebellar tract.

Reticulocerebellar mossy fibres stem from the lateral and paramedian reticular nuclei of the medulla oblongata (Ch. 21). The lateral reticular nucleus supplies major collateral projections to the cerebellar nuclei. Spinoreticular fibres terminate in a somatotopical pattern within the ventral lateral reticular nucleus, which projects bilaterally, mainly to the vermis. Spinoreticular fibres from the cervical cord overlap with collaterals from the rubrospinal tract and a projection from the cerebral cortex, and all terminate in the dorsal part of the nucleus, which projects to forelimb regions of the ipsilateral hemisphere. The cerebellar cortical projection of the paramedian reticular nucleus is very similar to that of the ventral lateral reticular nucleus.

Primary vestibulocerebellar mossy fibres enter the cerebellum with the ascending branch of the vestibular nerve, pass through the superior vestibular nucleus and juxtarestiform body, and terminate, mainly ipsilaterally, in the granular layer of the nodule, caudal part of the uvula, ventral part of the anterior lobe and bases of the deep fissures of the vermis (**Fig. 22.21A**). Secondary vestibulocerebellar mossy fibres arise from the superior vestibular nucleus and the caudal portions of the medial and inferior vestibular nuclei, and terminate bilaterally, not only in the same regions that receive primary vestibulocerebellar fibres, but also in the flocculus and the adjacent ventral paraflocculus (the accessory paraflocculus of the human cerebellum), which lack a primary vestibulocerebellar projection (**Fig. 22.21B**). Some of the mossy fibres from the medial and inferior vestibular nuclei are cholinergic.

CORTICOPONTOCEREBELLAR PROJECTION

The cerebral cortex is the largest single source of fibres that project to the pontine nuclei (**Fig. 22.22**). The fibres traverse the cerebral

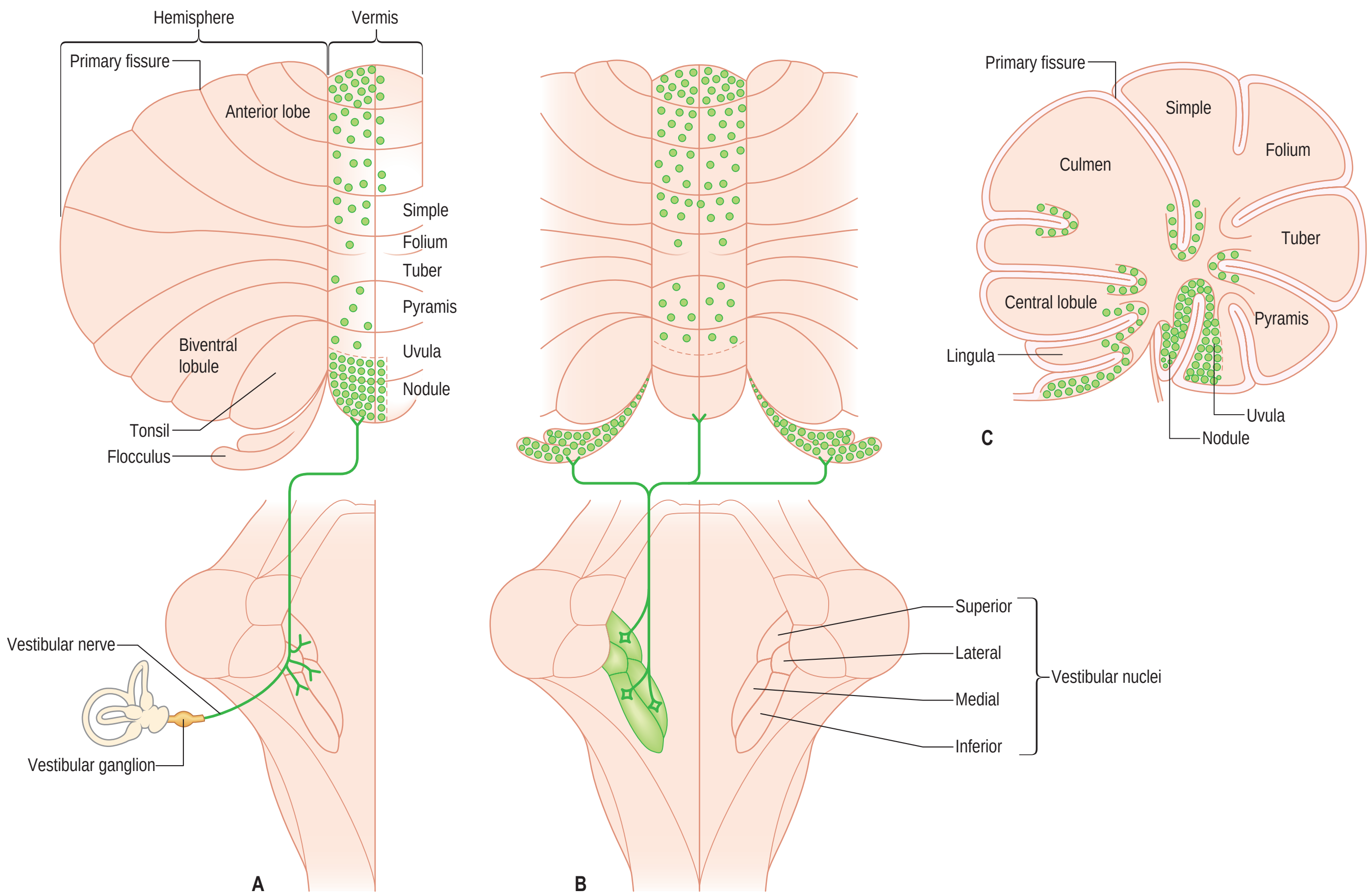

Fig. 22.21 Vestibulocerebellar mossy fibre projections. **A**, Primary vestibulocerebellar projections from the bipolar neurones of the vestibular ganglion. **B**, Secondary vestibulocerebellar projections from the vestibular nuclei. **C**, A sagittal section showing the distribution of both sets of afferents.

peduncle: those from the frontal lobe occupy the medial part of the peduncle; corticonuclear and corticospinal fibres occupy its central part; and fibres from the parietal, occipital and temporal lobes occupy its lateral part. The mediolateral sequence of the fibres in the cerebral peduncle is approximately maintained in their termination in the pontine nuclei. Prefronto-pontine fibres and the frontal eye fields project medially and rostrally; motor and premotor projections terminate centrally and caudally; and parietal, occipital and temporal fibres terminate in the lateral pontine nuclei (Schmahmann and Pandya 1997). Motor and premotor projections are somatotopically organized, such that the face is represented rostrally and the hindlimb caudally in the nuclei. In monkeys, the majority of the corticopontine fibres stem from motor, premotor and parietal areas. The prefrontal, general sensory and visual projections are relatively minor (Glickstein et al 1985). A prefrontal projection from the dorsal prefrontal cortex has been confirmed for humans (Beck 1950). Many corticopontine fibres are collaterals of axons that project to other targets in the brainstem and spinal cord (Ugolini and Kuypers 1986). The pontocerebellar projection is almost completely crossed. Fibres from the pontine nuclei access the cerebellum via the middle cerebellar peduncle and terminate throughout the entire cerebellar cortex, with the exception of lobule X (nodulus). Visual cortical mossy fibre input is found in the paraflocculus (tonsil, HIX). The pontocerebellar projection is still incompletely known; the relevant literature has been reviewed by Nieuwenhuys et al (2008) and by Voogd and Ruigrok (2012). **Figure 22.22C** is a simplified version of this projection (Glickstein et al 1985).

The nucleus reticularis tegmenti pontis (tegmental reticular nucleus of the pons) is located along the midline, dorsal to the pontine nuclei (see Fig. 22.22C). It gives rise to bilateral components of the middle cerebellar peduncle and receives a projection from the cerebellar nuclei via the crossed descending branch of the superior cerebellar peduncle. The medial, visuomotor, division of the nucleus reticularis tegmenti pontis receives visuomotor afferents from the frontal eye fields, the contralateral superior colliculus (the tectopontine tract) and other visuomotor centres in the brainstem, and targets lobule VII, the visual vermis and the adjacent crus I, the flocculus and the adjacent ventral paraflocculus. The bilateral projection of its lateral, motor, portion overlaps with similar projections from the pontine nuclei. Mossy fibres from the nucleus reticularis tegmenti pontis provide the cerebellar nuclei with a collateral innervation complementary to that of the lateral reticular nucleus. An uncrossed component of the tectopontine tract terminates in the dorsolateral corner of the pontine nuclei, where it overlaps extrastriatal visual afferents.

'OCULOMOTOR CEREBELLUM'

Traditionally, the flocculus and the nodulus are known as the 'vestibulocerebellum' because they maintain afferent and efferent connections with the vestibular system. They also belong to the functionally more comprehensive oculomotor division of the cerebellum, which includes lobule VII (visual vermis), the adjacent ansiform lobule, dorsal lobule IX (uvula), the ventral paraflocculus (the human accessory paraflocculus) and the dorsal paraflocculus (the human tonsil). The mossy fibre projection of the nucleus prepositus hypoglossi, a key element in the saccade-producing system (Ch. 41), outlines the entire oculomotor cerebellum, with the exception of the dorsal paraflocculus (Belknap and McCrea 1988) (**Fig. 22.23B**). The function of lobule X (nodulus) is not an exclusive oculomotor one because it also influences labyrinthine and postural reflexes and vegetative systems.

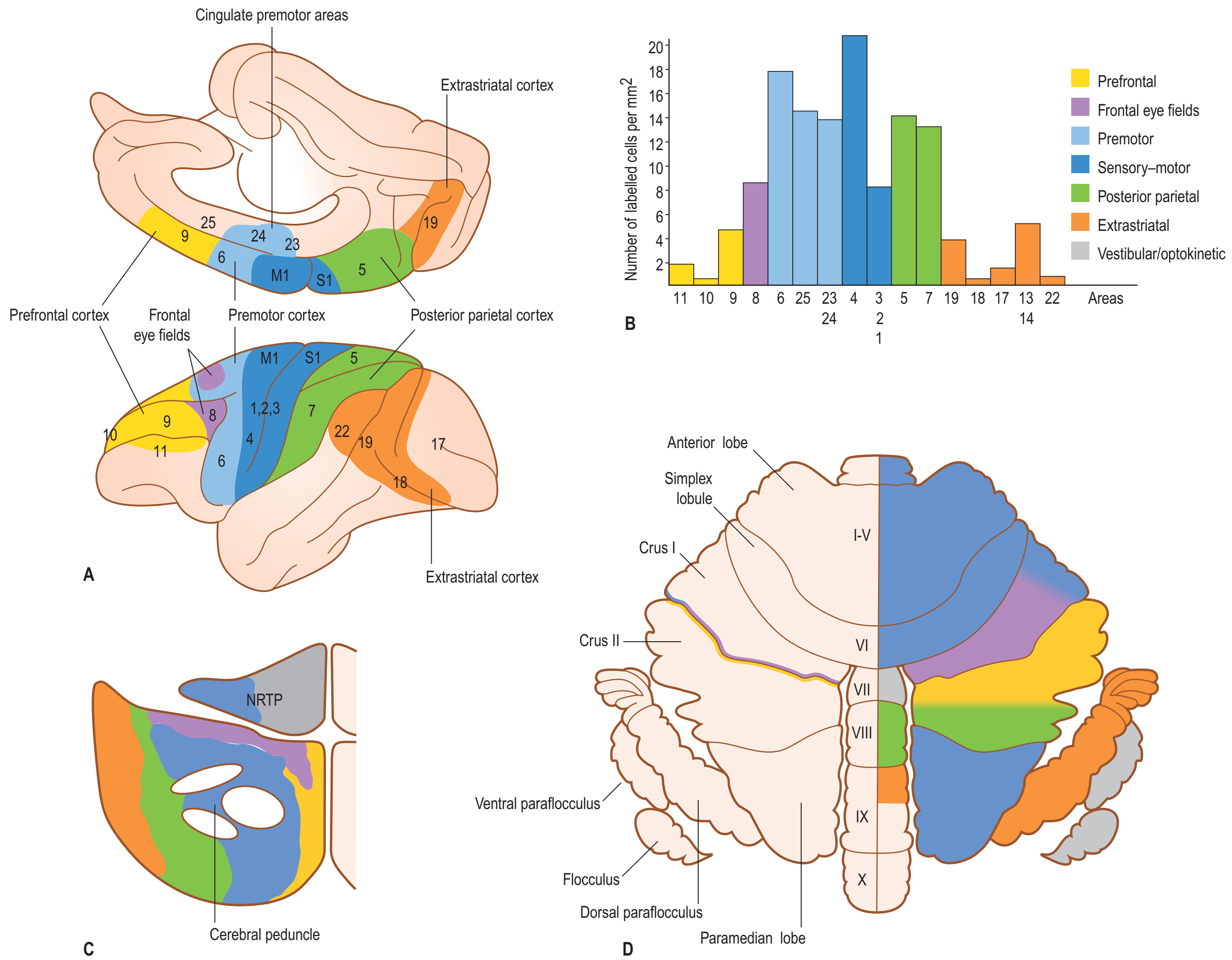

Fig. 22.22 The corticopontocerebellar system. **A**, The origin of corticopontine fibres from the cerebral cortex in the monkey (macaque). **B**, The relative proportions of corticopontine neurones in different areas of the cerebral cortex of the monkey, indicated in panel A. **C**, A transverse section through the pons showing the distribution of corticopontine fibres in the pontine nuclei and the nucleus reticularis tegmenti pontis (NRTP). **D**, The flattened cortex of the monkey cerebellum showing the distribution of pontocerebellar mossy fibres. (B, Modified from Glickstein M, May JG, 3rd, Mercier BE 1985 Corticopontine projection in the macaque: the distribution of labelled cortical cells after large injections of horseradish peroxidase in the pontine nuclei. J Comp Neurol 235:343–359.)

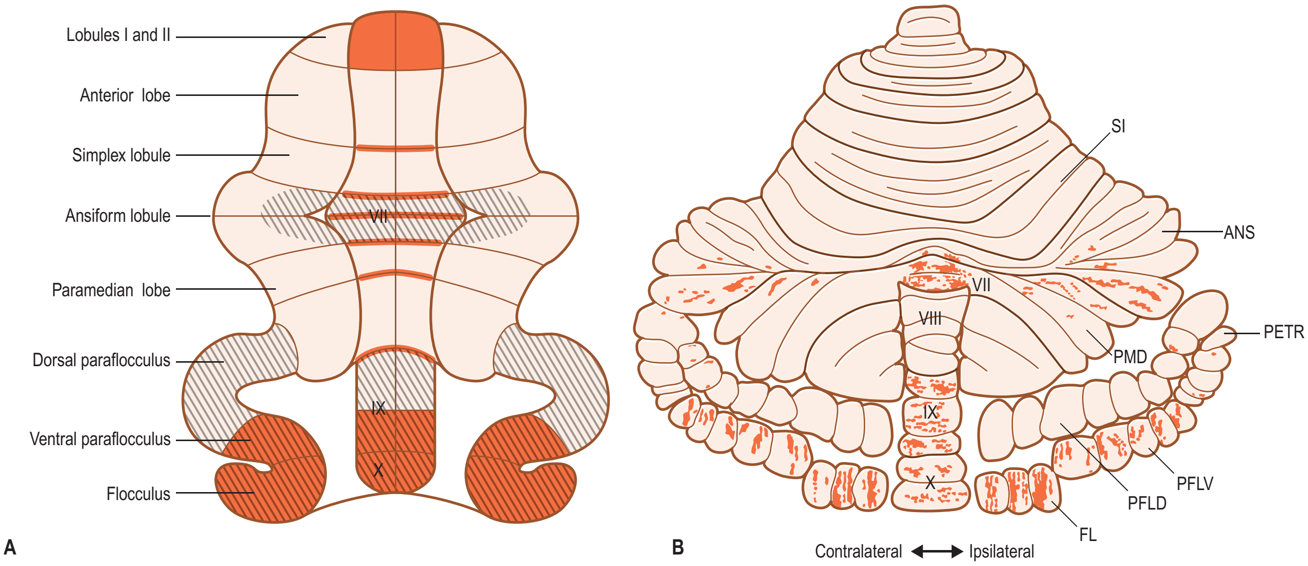

Fig. 22.23 **A**, A flattened map of the cerebellar cortex of the mammalian cerebellum showing the distribution of vestibulocerebellar mossy fibres in orange. The hatched lobules belong to the oculomotor cerebellum. **B**, The distribution of mossy fibres originating from the nucleus prepositus hypoglossi outlines the oculomotor cerebellum of the squirrel monkey, with the exception of the dorsal paraflocculus (PFLD). Other abbreviations: ANS, ansiform lobule; FL, flocculus; PETR, petrosal lobule; PFLV, ventral paraflocculus; PMD, paramedian lobule; SI, simplex lobule; VII–X, lobules VII–X. (Reproduced with permission from Belknap DB, McCrea RA 1988 Anatomical connections of the prepositus and abducens nuclei in the squirrel monkey. J Comp Neurol 268:13–28.)

The oculomotor cerebellum is involved in long-term adaptation of saccades, ocular stabilization reflexes and smooth pursuit (reviewed in Voogd et al (2012)). The role of the flocculus and the adjacent ventral paraflocculus in long-term adaptation of the vestibulo-ocular reflex (VOR) has been extensively studied. It is one of the few instances where the function of the cerebellum is clearly understood.

Vestibulo-ocular reflex

The VOR is an ancient reflex, being present in agnatha and fishes. It stabilizes the position of the retina in space, during movements of the head, by rotating the eyeball in the opposite direction. The VOR is an open reflex; there is no time for a feed-back connection that would compensate for inaccuracies in the execution of the reflex. This function is taken over by the long-term adaptation of the reflex by the flocculus. The circuitry of the flocculus, similar to the VOR, is organized on the coordinate system of the semicircular canals (Simpson and Graf 1981, van der Steen et al 1994). The VOR consists of different components. One component connects the lateral (horizontal) semicircular canal, via oculomotor neurones in the vestibular nuclei, with the oculogyric muscles that move the eyes in a plane co-linear with the plane of the lateral canal (**Fig. 22.24**). The anterior semicircular canal influences the ipsilateral superior oblique and the contralateral inferior oblique muscles that move the eye in the plane of this canal. (For further details, see Chs 38 and 41.)

Five Purkinje cell zones are present in the cortex of the flocculus and the adjacent ventral paraflocculus. Apart from the C2 zone, located most medially, two pairs of zones occupy its lateral portion. Zones F1 and F3 connect with the oculomotor neurones in the vestibular nuclei, subserving the anterior canal VOR. The F2 and F4 zones connect with oculomotor neurones of the horizontal canal VOR. The flocculus and ventral paraflocculus receive vestibular mossy fibre input, relaying an efferent copy of the output of the vestibulo-oculomotor neurones. They also receive climbing fibre input, signalling retinal slip that occurs when the stabilization of the retina by the VOR is incomplete. Retinal slip is perceived by two groups of neurones in the mesencephalon. In the horizontal plane, it is relayed by the nucleus of the optic tract. This nucleus, located in the pretectum, receives fibres of the contralateral optic nerve via the optic tract and projects to the dorsal cap of the inferior olive, located dorsomedial to the caudal medial accessory olive (see Fig. 22.15). The dorsal cap provides the F2 and F4 zones with climbing fibres. Retinal slip in the plane of the anterior canal is conveyed by the lateral and medial nuclei of the accessory system, which belong to a group of nuclei located on the periphery of the rostral mesencephalon, receiving optic nerve fibres from an offshoot of the optic nerve, known as the accessory optic tract. These nuclei project to the ventrolateral outgrowth of the inferior olive, located immediately rostral to the dorsal cap (see Fig. 22.15). The ventrolateral outgrowth innervates the F1 and F3 zones. Repeated simultaneous activation of the vestibular mossy fibre-parallel fibre input and the climbing fibres, relaying retinal slip, induces plastic changes in the Purkinje cell output that compensates for the retinal slip. Combinations of the horizontal and anterior canal systems ensure compensation of retinal slip in all possible planes. Knowledge of this system has been instrumental in the concept that climbing fibres are carriers of error signals, used in cerebellar learning (Marr 1969, Ito 1982).

NEUROIMAGING AND THE FUNCTIONAL DIVISIONS OF THE CEREBELLUM

Mossy fibre projections have been studied in the human brain using fMRI. In the cerebellum, activity in climbing fibres and Purkinje cells is overwhelmed by the massive activity of the mossy fibres (Diedrichsen et al 2010), which means that the modular organization of the cerebellum therefore cannot be visualized with this method. The division of the human cerebellum into anterior and posterior motor and intermediate non-motor portions has been observed in numerous neuroimaging studies (reviewed by Stoodley and Schmahmann (2009)). The somatotopical localization in each hemisphere of the anterior lobe and the simplex and biventral lobules in the posterior lobe has been confirmed with fMRI (Grodd et al 2001, Buckner et al 2011, Yeo et al 2011). A systematic somatotopical gradient has been reported for the digits of the hand in the hemisphere of lobule V (Wiestler et al 2011).

The crura of the ansiform lobule (HVII) are activated during the execution of cognitive tasks. More recently, resting-state functional connectivity fMRI has been used to map topographical correlations between remote, functionally coupled regions in the cerebral cortex and the cerebellum. Several functional networks in the cerebral cortex have been identified with this method (Yeo et al 2011) (**Fig. 22.25**). However, it does not provide information on the anatomical connections or the excitatory or inhibitory nature of the constituent areas of each of these systems; connections between the cerebrum and the cerebellum could be indirect, e.g. through cortical association systems or brainstem nuclei other than the pontine nuclei. The networks are distributed in a mirrored fashion in the anterior and posterior cerebellum. The default mode network, a network of brain regions that are active when an individual is not focused on the outside world (Buckner et al 2008; Commentary 3.1), occupies a central position.

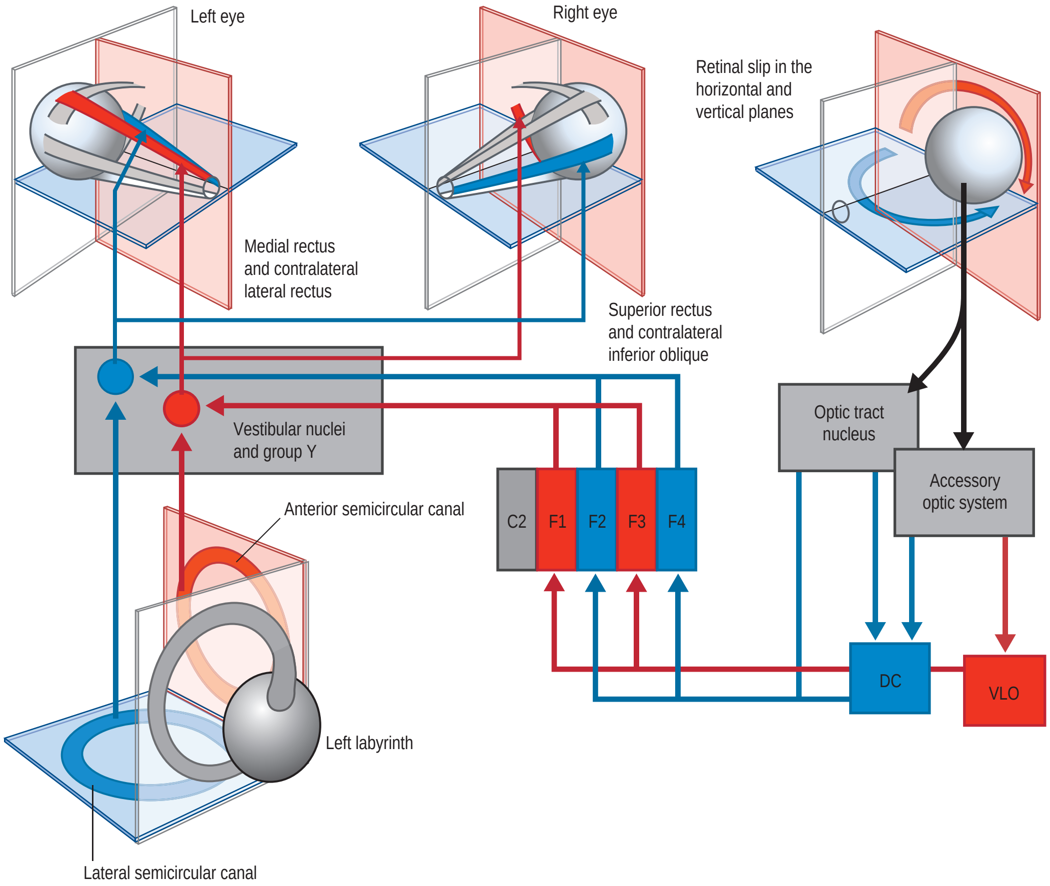

Fig. 22.24 The circuitry used by the flocculus in long-term adaptation of the vestibulo-ocular reflex (VOR). The system is organized in the planes of the semicircular canals. For an explanation, see the text. Abbreviations: C2, C2 Purkinje cell zone; DC, dorsal cap; F1–F4, floccular Purkinje cell zones F1–F4; VLO, ventrolateral outgrowth. (Modified with permission from Voogd J et al 2012 Visuomotor cerebellum in human and nonhuman primates. Cerebellum 11:392–410.)

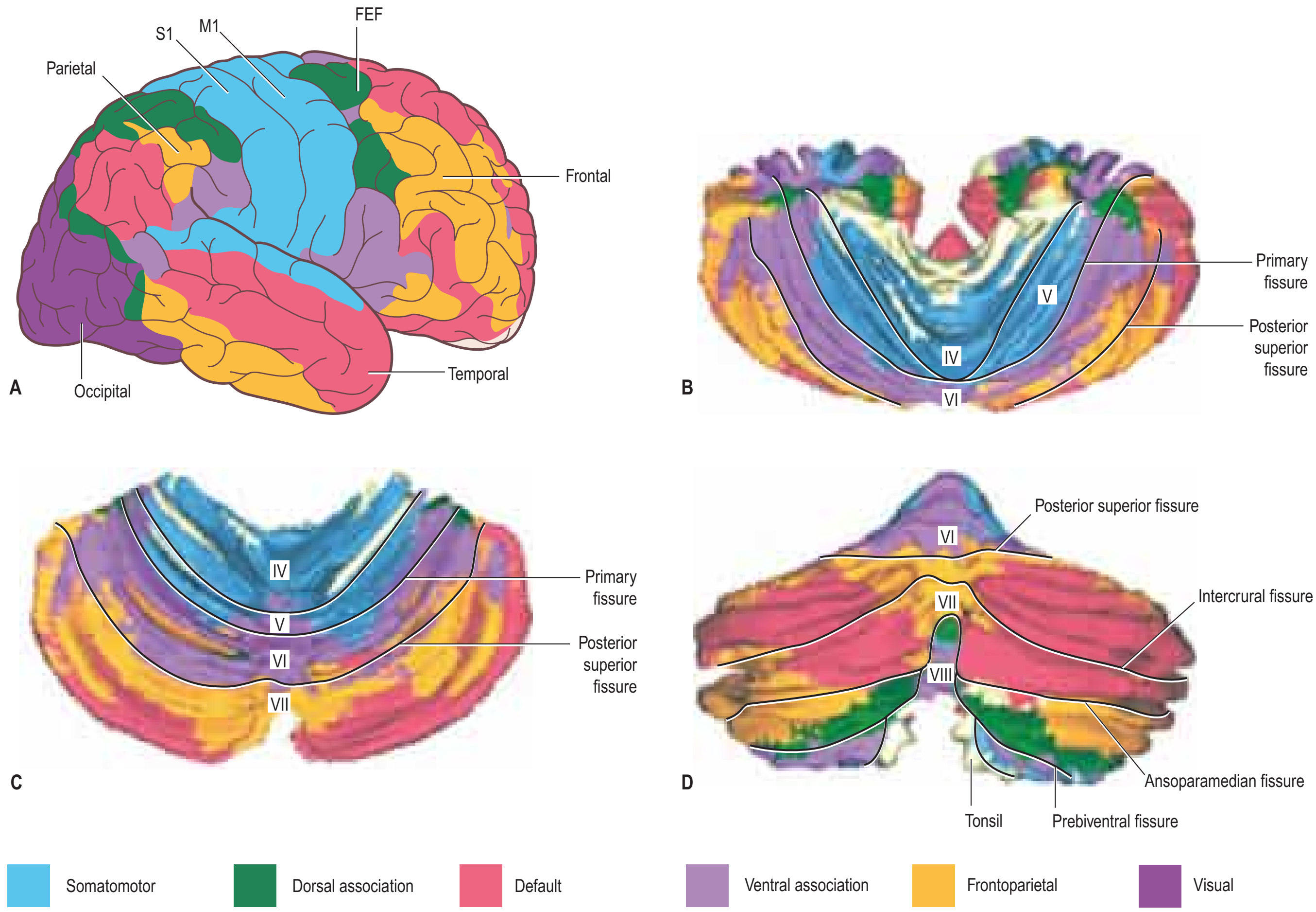

Fig. 22.25 A map of the topographical correlations between remote, functionally coupled regions in the human cerebral cortex and the cerebellum using resting state functional connectivity fMRI. **A**, Networks distinguished in the cerebral cortex. **B**, An anterior view of the human cerebellum showing regions that are functionally coupled to the different cerebral networks. **C**, A dorsal view of the human cerebellum. **D**, A caudal view of the human cerebellum. Abbreviation: FEF, frontal eye field. (A, Redrawn from Yeo BTT, Krienen FM, Sepulere J 2011 The organization of the human cerebral cortex estimated by intrinsic functional connectivity. J Neurophysiol 106:1125–1165; B–D, Reconstructions based on the transverse sections illustrated in Buckner RJ, Krienen FM, Castellanos A 2011 The organization of the human cerebellum estimated by functional connectivity. J Neurophysiol 106:2322–2345.)

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